TEC | Tecnológico de Costa Rica

Escuela de Ingeniería en Computación

Programa de Maestría en Computación

A Neural Network Framework for Estimation of Three Dimensional Object Location in Small Rooms Using Acoustic Echoes

A thesis submitted in partial fulfilment to opt for the degree of

Magister Scientiæ in Computer Science

Author
Esteban Roberto Padilla Padilla

Supervisor
Ing. Kervin Sanchez Herrera, M.Sc.

October 26, 2026

Thesis approval certificate goes in here.

To my wife Rebeca and my family, for the time they granted me, and for their unwavering support and patience throughout the writing of this thesis.

A mi esposa Rebeca y a mi familia, por el tiempo cedido, su apoyo y paciencia durante la realización de esta tesis.

Acknowledgements

This section can be used for the acknowledgements.

Abstract

This document presents the thesis for obtaining the Master's degree in Computer Science: Emphasis in Computer Science at the Tecnológico de Costa Rica. The topic is centered in the idea of the use of machine learning techniques to analyze the patterns in acoustic echoes produced inside small rooms to estimate the location of certain objects in a tridimensional space. Two neural networks architectures are employed for this task, a Convolutional Neural Networks that despite being an architecture mainly for images is commonly used in literature for acoustic inference, and an Echo State Machine as a new architecture with the potential to address in a more direct manner the acoustic signals. The effect of beamforming is also considered as one of the elements to investigate as a solution to the multi-path acoustic superposition. The design for the research is presented, the required experimental setup is discussed as well as the methodology to measure, process and evaluate the results, all this with the objective to evaluate the potential use of sound echoes in the audible frequency as a mean to locate object in an enclose room.

Contents

1. Introduction **1**

1.1. Problem 3

1.2. Document Structure 4

2. Theoretical Framework **5**

2.1. Room Impulse Response 5

2.2. Acoustic Beamforming 6

2.2.1. Microphone Array 7

2.2.2. Beamforming Pipe 8

2.2.2.1. Delay-and-Sum Beamforming Technique 8

2.2.2.2. Deconvolution 8

2.2.2.3. Post-Filtering 9

2.3. Time Series Transformation into Images 9

2.3.1. Markov transition Field 9

2.3.2. Gramian Angular Field 11

2.4. Convolutional Neural Network 11

2.4.1. Convolution operation 11
2.4.2. Pooling 13
2.4.3. Fully connected layer 14
2.5. Echo State Network 14
2.5.1. Formalism of ESN 15
2.6. Related Work 16
2.6.1. Passive Acoustic Sensing and Analysis 17
2.6.1.1. Acoustic Imaging 17
2.6.1.2. Acoustic Scene Classification 17
2.6.1.3. Acoustic Object Recognition 19
2.6.2. Active Acoustic Sensing and Analysis 20
2.6.2.1. RIR for Room Inference 20
2.6.2.2. Acoustic Object Shape Estimation 21

3. Hypothesis and Objectives **23**
3.1. Hypothesis 23
3.2. Objectives 24
3.2.1. Main Objective 24
3.2.2. Specific Objectives 24
3.2.3. Deliverables 25
3.3. Scope and Limitations 25

4. Methodology **27**
4.1. Experiment Design 27
4.2. Experimental measurement configuration 28
4.3. The Audio-Location Dataset for Model Training and Testing 29
4.4. The Audio-Location Processing Pipeline 31
4.5. Factors and Levels 33

4.6. Response Variables 34
4.7. Data Recollection for Model Evaluation 34
4.8. Statistical Tool Used for hipotesis Testing 35

5. Design **37**

5.1. CNN Estimator 38
5.1.1. Log-Mel spectrogram as images 38
5.1.2. CNN Architecture 40
5.1.3. CNN Training and Evaluation 40
5.2. ESN Estimator 41
5.2.1. Mel spectrogram as sequence 41
5.2.2. ESN Architecture 42
5.2.3. ESN Training and Evaluation 46
5.3. Beamforming Implementation 48
5.4. Beamforming 48
5.4.1. Frequency Selection 48
5.4.2. Audio compactation 51
5.4.3. Limitations and Requirements 51

6. Results **53**

6.1. CNN for Acoustic Classification 54
6.1.1. CNN Acoustic to Spatial Location Classification 54
6.1.2. CNN Acoustic to Orientation Classification 55
6.2. CNN Acoustic to Spatial Location Regression 59
6.3. ESN for Acoustic Classification 64
6.4. ESN as regression model 65
6.5. Beamforming 70
6.5.1. CNN with Beamforming 71

6.5.2. ESN with Beamforming 77
6.6. Correctness (if needed) 77
6.7. Output (If needed) 77

7. Discussion **79**
7.1. CNN model and the regression problem 80
7.1.1. CNN model to identify cube regions and orientations 80
7.1.2. CNN model to estimate the cube location 81
7.2. ESN model and the regression problem 84
7.3. Beamforming 86
7.3.1. CNN with Beamforming 86
7.3.2. ESN with Beamforming 86
7.4. Model Comparison 86

8. Conclusions and Future Work **87**
8.1. Conclusions 88
8.2. Future Work 88

9. Appendixes **89**
9.1. Source Code 90
9.2. Gathered Data from the Experiments 90

References **91**

List of Figures

2.1. Room Impulse Response and the geometrical room representation [31] . 6
2.2. Microphone array in binaural arrangement [43] 7
2.3. Summarized process to build a Markov Transition Field [13] 10
2.4. Convolution operation over a tensor of 5x5 with zero padding and a stride of 1, resulting in a festure map of 5x5 [47] 12
2.5. Pooling escalation with average pooling and max-pooling 13
2.6. Structure of Convolutional Neural Network for aircraft structural health monitoring [36] 14
2.7. Comparison of B) RC vrs A) traditional RNN, the bold arrows are weights that change during training [23] 15
2.8. Image on the left is an acoustic image, on the right and microphone camera used to capture the image [32] 17

4.1. Measure box: (a) Conceptual sketch for the acoustic measurement box, and (b) actual implementation for the data acquisition task. 29

4.2. Scatter plot of cube locations for each recording in the dataset, the blue columns represent the 26 sessions, and each point in the column represents the nine orientations captured at that location. 31
4.3. Acoustic to location processing pipeline 32

5.1. Log-Mel spectograms correspondig of the four channels of signals captured at session 1, location X=22.8, Y=-0.94 , Z=-9.43 and orientation 50 degrees. 39
5.2. Network accuracy vs. reservoir size for 50 epochs runs. 43
5.3. Test MAE as a function of Intermediate Hidden Layer Size for different Sparsity values 45
5.4. Multihead Network architecture for the ESN Estimator. 47
5.5. Frequency response of the stimulus signal showing the top 10 spectral components with the highest magnitude. 49
5.6. Geometric layout of the measurement box illustrating the acoustic path from the speaker located at E to the microphones positioned at G and H. 51
5.7. Acoustic waveforms at 400 Hz captured by the microphones located at points G and H. 52

6.1. Confusion matrix for CNN acoustic location classification on test set. . 54
6.2. Confusion matrix for CNN acoustic orientation classification on test set. 55
6.3. Confusion matrix for the three regions defined for orientation classification. 56
6.4. Session point used for cross-correlation analysis. the image also shows the three regions for analysis of the orientation classification task. . . . 57
6.5. Cross-correlation results for distance and orientation for similarity analysis of the acoustic signal. 58
6.6. Histograms of estimation errors with CNN for X values across five folds. 59
6.7. Histograms of estimation errors with CNN for Y values across five folds. 60

6.8. Histograms of estimation errors with CNN for Z values across five folds. 60
6.9. 3D scatered plot showing the absolute error value for all the points estimated in the five folds as test set. 63
6.10. Scatered plot showing the absolute estimation error vs it respective coordinate axis. 64
6.11. Confusion matrix for the ESN as result of the audio session classification. 65
6.12. Histograms of estimation errors with ESN for X values across five folds. 66
6.13. Histograms of estimation errors with ESN for Y values across five folds. 66
6.14. Histograms of estimation errors with ESN for Z values across five folds. 67
6.15. 3D scatered plot showing the absolute error value for all the points estimated with the ESN in the five folds as test set. 68
6.16. Scatered plot showing the absolute estimation error vs its respective coordinate axis for the ESN. 70
6.17. Histograms of estimation errors with Beamforming and CNN for X values across five folds. 71
6.18. Histograms of estimation errors with Beamforming and CNN for Y values across five folds. 72
6.19. Histograms of estimation errors with Beamforming and CNN for Z values across five folds. 73
6.20. 3D scattered plot showing the absolute error value for all the points estimated in the five folds as test set. 75
6.21. Scattered plot showing the absolute estimation error vs the respective coordinate axis. 76

List of Tables

4.1. Factors and levels for the experiment. 33

4.2. Five-fold cross-validation structure and performance summary. 35

5.1. Key parameters of an Echo State Network (ESN) 42

5.2. Performance Across Intermediate Steps 44

5.3. Echo State Network (ESN) Hyperparameters 44

5.4. Error Distribution Comparison Between Standard ESN and MultiHead ESN 45

5.5. Top 10 frequencies with the highest magnitude (dB). 49

6.1. Model Performance Across The Three Regions 56

6.2. Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates 61

6.3. Pearson Correlation Coefficients between Ground Truths and Predictions 61

6.4. Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates 62

6.5. ESN Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates 67

6.6. Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates 68
6.7. Pearson Correlation Coefficients between Ground Truths and Predictions 69
6.8. Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates 73
6.9. Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates 74
6.10. Pearson Correlation Coefficients between Ground Truths and Predictions 76

Chapter 1

Introduction

Robots are expected to navigate in multiples and different environments, in order to achieve this we need to allow them to have situational awareness and autonomy to make proper decisions. Computer vision aims to enable machines to get an interpretation of the world that surround them. Processing images and videos robots can extract information to recognize objects geometries, textures, materials, etc. Although it is true that with visual analysis, the machine can create internal models that contain a vast knowledge of the environment, this doesn't mean that its perception is complete. Robotic perception give the ability to a machine to understand its surroundings to the degree that could safely interact with the world [15], for such understanding other techniques are required as well as computer vision to get a complete sensory experience of the environment. Acoustic sensing and analysis can enhance the perception of machinery following an approach similar to that seen in nature with animals that use binaural acoustic perception, the environment can be analyzed to detect the source of a particular sound and focus the visual attention to that particular region, but, this capability is not only an extension,

in some conditions the animal can entirely relies on the acoustic information to operate, this is the case of echolocation employed by bats, birds, dolphins and wales, they ear the echoes of acoustic chirps to gain spatial perception of the surrounding environment and this enable them to navigate in complex layouts [45].

Traditional computer vision techniques that employ sensing devices such as cameras and lidars provide high-fidelity information to navigation systems, but they become unreliable in scenarios with poor light or presence dust, smoke, and fog [37]. Another aspect to consider is the cost of the sensing device, here the acoustic perception offers an advantage due the low cost of commercial electret or MEMS microphones which are inexpensive compared to optical cameras and lidars; then become desirable its use as an alternative to traditional sensors. Recent developments in machine learning as the ones shows in [5; 30; 37] demonstrated the big potential to use neural network to infer spatial and geometrical information of objects in the surrounds by just analyzing acoustic echoes in the audible spectrum. The manufacturing sector under the industry 4.0 transformation is increasing in the adoption of automation and smart monitoring technologies and so it has become an important field of application of these new methods, where the supervision of machinery during operation, especially the machines allocated in production floors using robotic arms and computer numerical control actuators intended to manipulate objects in confined spaces [27]. In such conditions the acoustic sensing is a good approach to get an insight of the overall status of the system due to the nature of the Room Impulse Response as a transfer function that comprehends information of the environment setup. The acoustic response of a room can be use together with artificial neural networks like convolucional neural networks or recurrent neural networks to learn from this response and be capable of recognize characteristics and properties of the room echoes to recognize spaces and objects in it. Thus, acoustic perception together with neural networks can improve the perception of machinery

when it operates in complex spaces.

1.1. Problem

The use of acoustic signal to approximate the size and shape of objects was considerably improved by the adoption of machine learning techniques. An important part of the work in this field was develop with the scope of indoor navigation systems where the objective is to enhance the perception of robots using acoustic analysis to build low resolution deep images of an unknown environment. A different point of view is formulated and presented in this work, one in which we know the environment completely a priory, but we have an object that will change it's position inside the environment and we expect to know it's location with certain accuracy each time. Second, contrary to the large size room in which those systems are intended to navigate, we expect to estimate the object location in short ranges in which multi-path echoes could be received at the same time creating a superposition difficult to resolve and cause malfunction in those systems [5]. It is our understanding that there are no works related to estimating the position of an object through acoustic analysis under the circumstances described. To achieve the acoustic spatial estimation in this scenario we propose two key steps to the processing pipeline, one in the beamforming stage to increase the spatial discrimination from where the echoes are originated, and second a change in the type of neural network use to infer the location, to employ a network suitable to directly process time series instead of Convolutional Neural Networks. Our purpose is to demonstrate that is possible to locate an object in a small room (e.g. 1m side dimension) using acoustic signal processing with neural networks.

1.2. Document Structure

This document present in **Chapter 1** a brief introduction to the topic and is stated the problem we intent to address with the development of this thesis. **Chapter 2** contains the theoretical background and state of the art in the field. **Chapter 3** details the hypothesis, objectives and scope of the work. **Chapter 4** shows the research proposal and methodology. **Chapter 5** present the schedule for the execution of the proposed work.

Chapter 2

Theoretical Framework

In this chapter is described the theoretical concepts needed to understand the work that will be developed during the research's execution.

2.1. Room Impulse Response

An acoustic impulse is sharp and narrow pulse which spectrum is composed by a uniform distribution of all frequencies. The mathematics Dirac delta function $\delta(t)$ is used to represent this kind of impulse. In enclosed spaces the sound waves generated by an impulse will travel and reflect and refract in the wall delimiting the space, for a microphone measuring the environment, it will receive the direct path sound, the sound that travel from the source directly to the microphone, followed by the early reflections and finally a series of pulses called late field reverberation as show in figure 2.1 . The direct path, early reflection and reverberations correspond to the response of the enclosed space to and acoustic impulse and is physically defined by the environmental geometry and materials [34].

The Room Impulse Response (RIR) by definition is the transfer function of a sound wave that travel in the space from it's source to the microphone. The microphones hear the convolution of the emitted sound with the corresponding RIR [11] mathematically showed in equation 1, therefore any sound generated in a point and received in other inside a room is the composition of the original signal plus the RIR of the enviroment.

$$g(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(t)x(t-s)ds \tag{1}$$

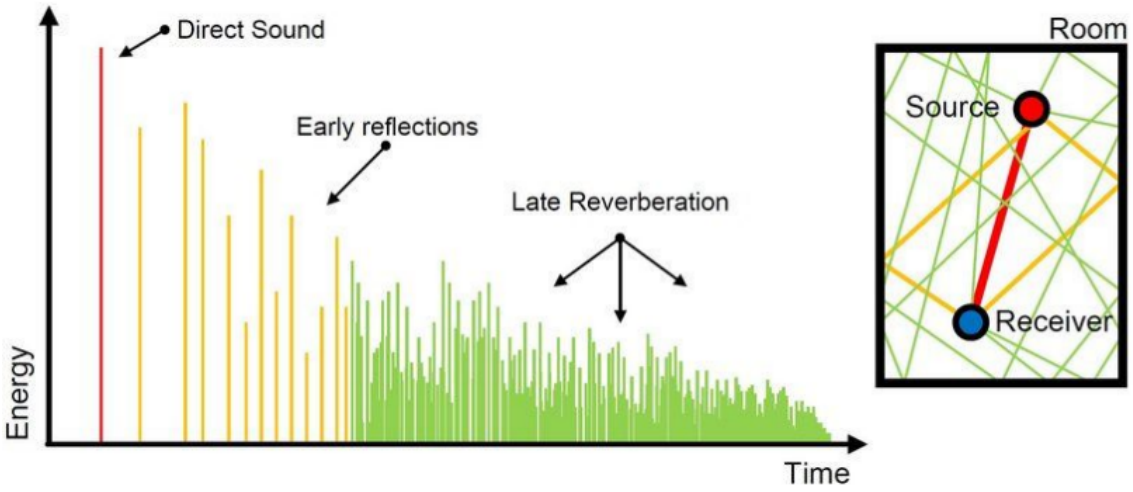


Figure 2.1: Room Impulse Response and the geometrical room representation [31]

2.2. Acoustic Beamforming

In many spatial omnidirectional data transfer context, not only the information carried in the medium is important, the source from where the signal is generated also provide valuable information to properly analyze the incoming data. Beamforming is a spatial filtering technique aims to estimate the direction of the arrival signal inclusive in the presence of noise and other interference in the propagation medium [8]. In acoustic

an array of microphones is used to capture the spatially propagated signals arriving from a certain direction and processes them to investigate the acoustic sources [26].

2.2.1. Microphone Array

A Microphone array combine the signal of multiple omnidirectional microphones in such a way that capture sound from a specific direction while reject sound from other directions [43]. The shape and dimensionality of arrangement depend on the application. The simplest form consist in two microphones spaced by a distance D as show in figure 2.2. The value for D is the minimum distance for a given frequency and its calculation is shown in the equation 2, in the case that the input signals are simple added together for a given frequency f, this arrangement capture sound coming from ± 90 degrees and rejects the sound at 0 and 180 degrees. In a microphone array the higher the number of microphones results in better gain in the desired direction as well as the attenuation of sound coming from the sides of the array [33].

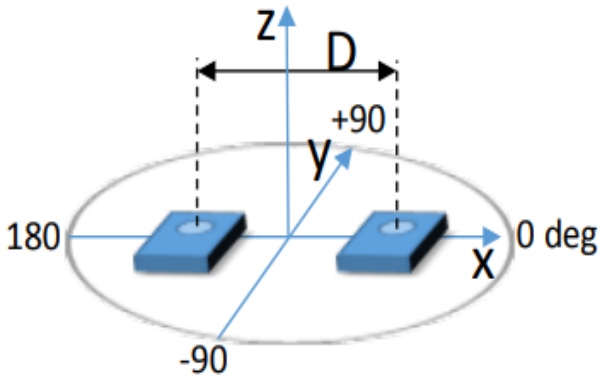


Figure 2.2: Microphone array in binaural arrangement [43]

$$D = \frac{1}{4}\frac{c}{f} \tag{2}$$

2.2.2. Beamforming Pipe

2.2.2.1. Delay-and-Sum Beamforming Technique

Beamforming is based on the physical principle that an acoustic wave originated on a particular location in the space, it travels and arrive at each sensor at different times defined by the distance existing between the unique source and each sensor. The result of adding all these signals depends of the phase relationship between them, so is possible to control the delay for each signal to constructive add the ones from a particular location and, at the same time, destructively cancel the ones originated in other regions [8]. Mathematically the beamforming formula is show in equation 3, where $L(t, S_0)$ is the beamforming output, pm is the microphone output for a signal originated at $S_0 = (x_0, y_0, z_0)$ and captured by a microphone located at S. All the signals are sum together and divided by the total number of microphones M in the array.

$$L(t, s_0) = \frac{4\pi}{M} \sum_{m=1}^{M} pm(s_0, t + t_0)|S - S_0| \tag{3}$$

2.2.2.2. Deconvolution

In real applications the beamforming output comprehends the acoustic signal convolved with the microphone array response. The deconvolution principle assumes that the recorded signal $r(t)$ is a convolution of the sound $g(t)$ (The result of equation 1) and a transfer function $ma(t)$ which represents the array response to a given source [8]. The Point-Spread-Function (PSF) is the array response to a monopole placed in front of the array. The PSF provides the information required to obtain the array transfer function required to do deconvolution and extract the sound without the distortion created by the array.

2.2.2.3. Post-Filtering

Taking into consideration that the signal measured for each microphone is the sum of the sound and the level of noise, and also that the noises are uncorrelated between each microphone and between the beamforming output signal. The beamforming output can be further enhanced by selectively filter the resulting signal based on the cross-spectral densities between channels in the microphone array [26].

2.3. Time Series Transformation into Images

Given a time series, for the sake of analysis sometimes is desirable to perform certain kind of transformations to the time series that provide a more compact representation of the data or highlight specif properties and attributes of it. Transforming time series into images involves the process of transforming vectors into matrices using a criterion of transformation that map values from one domain to another.

2.3.1. Markov transition Field

In the Markov trasnsition fields (MTF), given a time series $X = (x_1, ..., x_n)$ of real observations, the serie is discretized based on quantile bins assignation. First the codomine of the function is sub-grouped taking the range of values in X and divided it into Q segments, then each x_i is allocated to bin q_j with $j \in \{1, ..., Q\}$ if its value is within the segment. Now considering the resulting bin values as observations of a first-order Markov chain and computing the number of occurrences of pairs of back-to-back bins for every pair of bins is possible to represent the ocurrences as a Markov transition matrix of dimension $Q \times Q$. The Markov transition matrix is normalized to transform frequencies into probabilities. then each entry in the matrix is the probability to go from q_j to q_k, for every pair (q_j, q_k) [13].

Finally the dimensionality of the time series is restored by projecting the Markov

matrix $Q \times Q$ into a $n \times n$ matrix called Markov transition field, where each entry for the pair (x_i, x_j) is the probability to go from the bin $q(x_i)$ to bin $q(x_j)$. For a serie of n observations the MTF can formally be defined by the equation 4.

$$\forall i, j \in 1, ..., n, MTF_{i,j} = \mathbb{P}(q(x_i)|q(x_j)) \tag{4}$$

The figure 2.3 Shows the data trasformations requiered to obtain a MTF, First in (a) the time series is discretized by dividing the variable range in intervals, (b) the Markov transition matrix is calculated and from this the MTF for the probability of each pair in the serie is created as presented in (c).

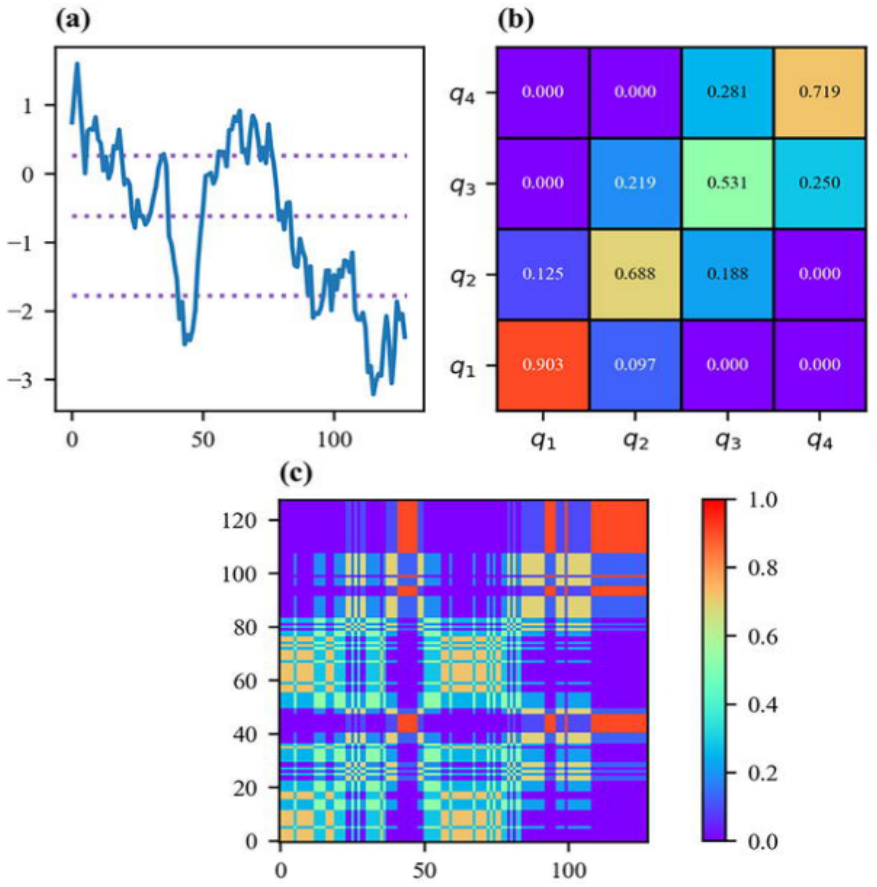


Figure 2.3: Summarized process to build a Markov Transition Field [13]

2.3.2. Gramian Angular Field

The Gramian Angular Field (GAF) is a matrix build on the principle of transform the time series values to the polar domain [41]. Given a time series $X = (x_1, ..., x_n)$ of real observations, the gramian transformation takes the values x_i and rescale them in the interval $[-1, 1]$ or $[0, 1]$. Then the rescaled values are converted to a polar form considering the rescaled value as the angular cosine and the timestamp the radius. The transformation function is show in 4. From the resulting values only the angle is used as the one that depends of the values of the time series. Two matrices can be build considering the sum or subtraction of pair of angles resulting from 4, the sum is called Gramian Angular Sum Field (GASF), the subtraction of angles receive the name of Gramian Angular Difference Field [14].

$$\begin{aligned} \phi_i &= \arccos(\bar{x}_i), \quad \bar{x}_i \in \bar{X} \\ r_i &= \frac{t_i}{n}, \qquad\qquad t_i \in \mathbb{N} \end{aligned} \tag{4}$$

2.4. Convolutional Neural Network

A Convolutional Neural Network (CNN) is an artificial neural network architecture used to analyze and learn from visual data presented commonly in the form of images. In its simplest form the structure is composed by three building blocks: convolution layer, pooling layer, and fully connected layer. The figure 2.6 shows and example of how this blocks are connected to analyze an image stage by stages in order to determine the structural condition of an aircraft fuselage.

2.4.1. Convolution operation

Convolution is a common operation used on image filtering to denoise or detect features. A filter also called a convolution kernel is applied to all locations of the image to produce a new filtered image [10]. For a given 2D image X of dimension $I \times J$ with

pixels $x_{i,j}$ a filter W of dimension $K \times L$ with weights $w_{k,l}$ can be convolved to obtain a filtered image $X * W$. The convolution process over the image can be mathematically written as:

$$(w * x)_{i,j} = \sum_{k=-K/2}^{K/2} \sum_{l=-L/2}^{L/2} w_{k,l} x_{i+k,j+l} \tag{5}$$

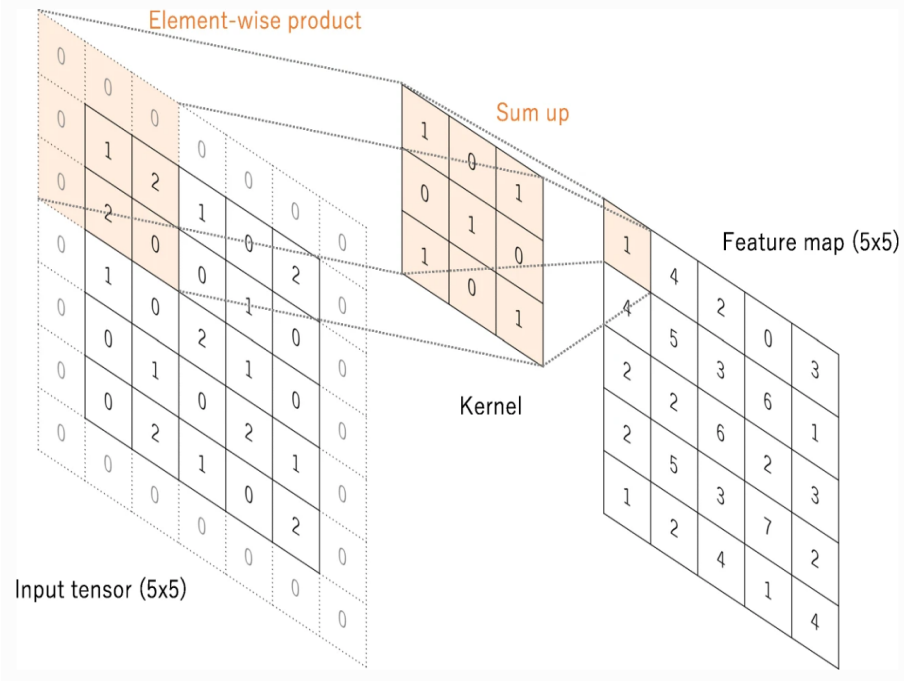


Figure 2.4: Convolution operation over a tensor of 5x5 with zero padding and a stride of 1, resulting in a festure map of 5x5 [47]

The convolution operation in equation 5 does not allow the center of the kernel to overlap the edges of the image, resulting in a reduction in the size of the output feature map. Padding increase the input size by adding extra rows and colums arround the outermost elements on each side of the image in order to compensate the dimensional reduction [47]. In the practice as show in figure 2.4 the kernel is applied across the input one step at the time, an element wise operation is perform in this

location and summed to obtain the output, after this the kernel is moved to the next location to filter other segment of the input. How big is displacement of the kernel is given by the stride, which define the distance between two succesive kernel positions.

2.4.2. Pooling

Pooling is a dimension reduction technique where an input map is spatially reduced while keeping the most important features filtered or learned. The pooling operation reduce or collapse a sub region to a smaller dimension helping the learning representation to be more invariant to small length-scale perturbations [19].

Two common pooling techniques are max-pooling and average-pooling, the first one, for the defined pooling window it will consider only the maximum value,in the case of the average pooling is calculated the average of the values inside the window as show in the example of figure 2.5.

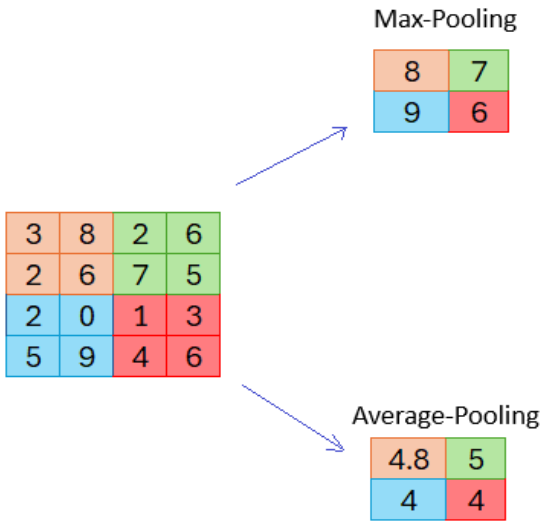


Figure 2.5: Pooling escalation with average pooling and max-pooling

2.4.3. Fully connected layer

The final stage in a CNN is a fully connected layer. This layer is the responsible to classify the features extracted from the previous layers. The figure 2.6 shows and example of CNN for aircraft structural health classification with it fully connected layer allocated at the output of the net.

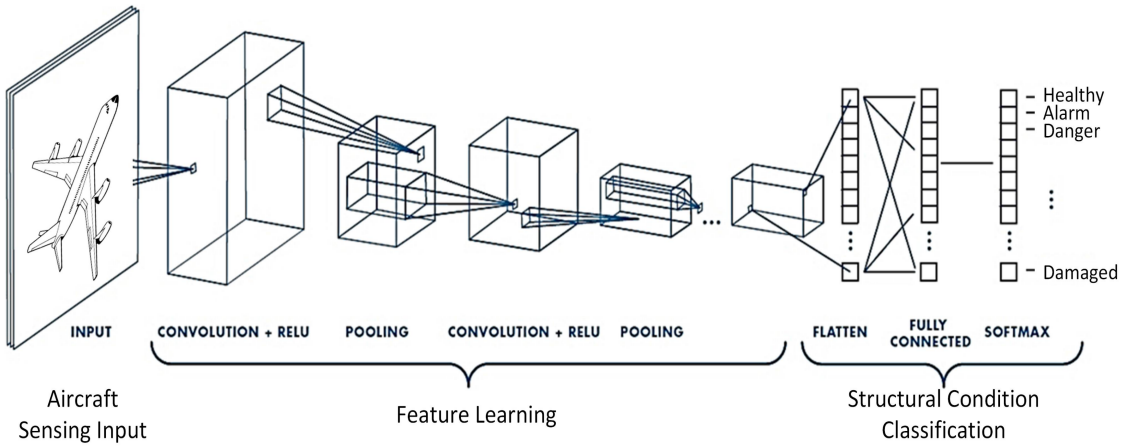


Figure 2.6: Structure of Convolutional Neural Network for aircraft structural health monitoring [36]

2.5. Echo State Network

An Echo State Network (ESN) is a new paradigm in recurrent neural networks called Reservoir Computing (RC). In RC, the reservoir is the recurrent part of the network and has the particularity that is not trained. Due the fact that the reservoir skip the training the network avoids the computationally expensive of gradient-descendant on recurrent networks, and instead of training the recurrent network is defined is such a way that is capable of transform input signals into a high-dimensional state space values. The state of the reservoir can be process by a simple fully connected read out network trained to map the reservoir state to a desire output [23].

In the ESN, the reservoir is a randomly created RNN where the weights remains

unchanged. This RNN is exited by an input signal, and its future state is a function of the actual input and the history of the reservoir. The result is generated by training a simple feed-forward network to output the target value. Figure 2.7 [23] shows a comparison of traditional RNN and RC. In RC the bold arrows used to exemplify the weight change resulting from the training process and is only performed at the output layer of the network.

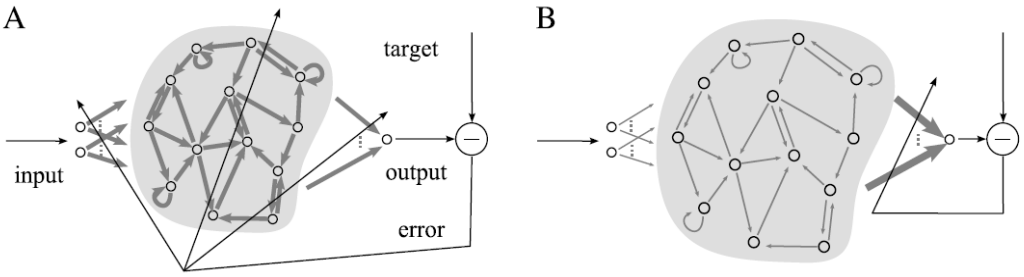


Figure 2.7: Comparison of B) RC vrs A) traditional RNN, the bold arrows are weights that change during training [23]

2.5.1. Formalism of ESN

A formal description of an ESN from [35] define an ESN receiving a time series input $U(t) \in R^{D\times 1}$, the state of the reservoir is denoted by $X(t) \in R^{N\times 1}$ and $y(t) \in R^{M\times 1}$ the output. The weights for the stages are: $W_{in} \in R^{N\times D}$ for the input, $W_{res} \in R^{N\times N}$ for the reservoir and $W_{out} \in R^{N\times M}$ for the output. The equations for the internal state of the reservoir and the output are show in 6. The reservoir state depends of the input value at time t and the previous state of the system. The output can be obtained by training to solve a linear regression problem.

$$\begin{aligned} x(t) &= f(W_{in}U(t) + W_{res}x(t-1)) \\ y(t) &= W_{out}x(t) \end{aligned} \tag{6}$$

There exists an important property that the reservoir should meet to ensure the reservoir dynamics is capable of forget the initial transient state [46]. The Echo State Property (ESP) enunciate that a network with state transition equation F has the echo state property if for each input sequence $U = [u(1), ...u(N)]$ and all couples of initial states x, x' for the reservoir, it hold the equation 7.

$$||F(U, x) - F(U, x')|| \to 0 \; as \; N \to \infty \tag{7}$$

To ensure the existence of ESP the W_{res} must meet a necessary condition and a sufficient condition as defined as follow:

$$\begin{aligned} ESP &\Rightarrow \rho(W_{res}) < 1 \\ \bar{\sigma}(W_{res}) < 1 &\Rightarrow ESP \end{aligned} \tag{8}$$

Where $\rho(W_{res})$ is the spectral radious parameter of W_{res} and the $\bar{\sigma}(W_{res})$ is the largest singular value of W_{res}. So from equations 8, if $\rho(W_{res})$ and $\bar{\sigma}(W_{res})$ are less than one, the ESP is satisfied and the reservoir dynamic will be correct.

2.6. Related Work

The related work in acoustic processing for object recognition and classification can be divided in two categories based on from where the sound is originated. We can considered a passive sensing approach if the sound stimulus is originated directly in the system, otherwise if the sound is an external stimulus that interact with the

environment before sensed we are under a active sensing approach.

2.6.1. Passive Acoustic Sensing and Analysis

2.6.1.1. Acoustic Imaging

An acoustic image is a sound intensity map that highlight the location of different sound sources in a bi-dimensional plane. An acoustic camera takes this intensity map and superimposed it on a optical image to show the physical source of the sound. As show in figure 2.8 a microphone array is used to ear the sound and a beamforming algorithm process the signals to locate the direction from where the sound is originated.

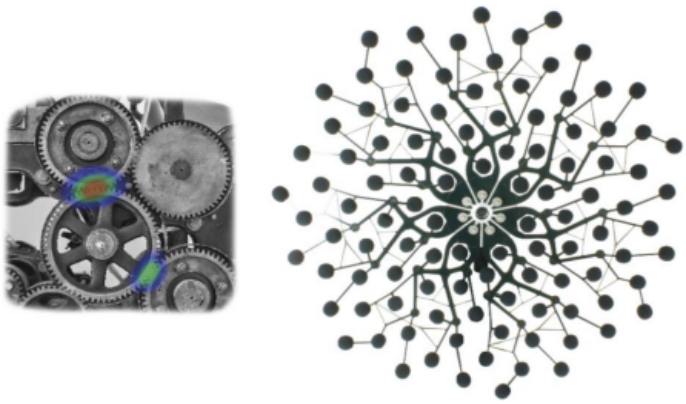


Figure 2.8: Image on the left is an acoustic image, on the right and microphone camera used to capture the image [32]

Acoustic beamforming is a field that has been extensively studied, different shapes and sizes of microphone arrays has been analyzed [33], also exist many techniques to beamforming the signals from the array [26], recently studies are focused on spherical arrays to identify multiple sound sources in a three dimensional space [9; 17; 21].

2.6.1.2. Acoustic Scene Classification

Artificial Intelligence and specially the deep learning subfield powered the development of new techniques to segment, analyze and classify acoustic originated

signals. Acoustic Scene Classification (ASC) use deep learning to learn and distinguish from audio the context in which the recording was performed. The CNN are the first choice for ASC, the required images that are going to be the input to the network are obtained by processing an audio dataset to convert it to spectrograms. Taking the DCASE 2016 dataset, the static Log-Mel spectrograms and dynamic first derivative spectrograms are used in [7] to build the two channels of the images that will be used as input training of the CNN. Han et al. in [16] used the DCASE 2017 dataset to improve the input data to train, this through multiple versions of mel-spectrograms divided in binaural representation, source separation and background subtraction. In binaural representation instead of mono audio representation they used left-right and mid-side spectrograms. The source separations divide the dataset into two using harmonic-persecutive sound separation algorithms to increase the dataset to get the representation in armonic and percusive sound. Finally media filtering is used in the spectrograms to steady noise from the environment. The resulting multiple version spectrograms are used to training a ConvNet architecture of 8 convolution layers inspired in a VGG net. They obtained a mean accuracy improvement compared with the baseline used in the work and a final 74.8% to a 80.5% accuracy with the various prepossessing methods previously described. The work presented in [39] used the DCASE 2019 dataset and the log-mel spectrograms to training two deep learning algorithms to perform the classification, a CNN but also a Convolutional Recurrent Neural Network (CRNN) are the architectures choosed in this work. The highest recognition obtained was for the CRNN with 90.96% of accuracy, a value superior to 73.61% of the CNN used as baseline. A different objective is presented by Byttebier in [6] where small foodprint neural network models are investigated. The author compared several small neural networks employing the DCASE 2020 to evaluate them and found that a 16-bit CNN with squeeze excitation blocks, regularized with MIX UP and weight decay is the one that shows the best

accuracy of 66.84% with only 90 kilobyte of parameters.

2.6.1.3. Acoustic Object Recognition

Acoustic object recognition aims to recognize an object in base of the sound that emits whether it is self-generated as the sound produced by and air-conditioner during normal operation or external, in response to a stimulus like the characteristic sound produced by the wood when someone is knocking a door. For the work presented by Luo et al. in [24] a dataset was built using 30 different objects, each of them was hit by a marker pen to generate a characteristic sound that was recorded. A training set of 100 hits (in different locations of the object) per object was created, and an extra 20 hits per object were obtained as well in order to be use in the test phase. With the dataset a denoise-autoencoder was trained to recognize the different sound of the object from which its originated. The recognition rate for the autoendoders was 91.50%, a value superior to the baseline provided by a Support Vector Machine with recognition rate of 82%.

Other dataset was created by Tran et al. in [38], in this case the input were LEGO objects that fall from 30cm high. The sound of the LEGO pieces hitting the ground was used to create spectrograms to training a CNN called LegoNet in order to recognize the LEGO pieces by the sound of its free-fall impact, LegoNet achieved an accuracy of 99.21%, a value superior to the other networks used for comparison. For robotic perception in [20] was develop a framework in which a robot can recognize the object inside a container by shaking the container and analyzing the sound. An kernel K-nearest neighbor algorithm in a open environment is used to shows the feasibility of the method. A generative physical, audio-visual synthesis pipeline is presented in [48], the model takes as input the shape of an object, its material and one scene configuration, a physical engine simulate motion and collision for that particular object. An audio engine takes the shape, material and the collision information to synthesize the sound.

The database generated is used in the training a two-stream CNN, one ResNet-18 for the image data, together with a SoundNet-8 for the audio data. The net demonstrated that the knowledge learned in the synthetic audio-visual dataset was transferred to real world data once it was used to infer object materials from auditory data, and estimate shape attributes from audio-visual data.

2.6.2. Active Acoustic Sensing and Analysis

2.6.2.1. RIR for Room Inference

For an enclosed space, the RIR provided a unique representation of the geometry and material inside it, sound like music and speech recorded in rooms comprehend this echoes information and can be use to recognize the room in which the recording was performed. This idea is presented in [28] where the authors developed a machine learning system to identify rooms for the reverberant audio samples recorded in those rooms and achieved and 85% accuracy in the task. The work in [4] use the RIR and speech data to train a model to infer the room volume from a single microphone recorded sound, with this information the model with an accuracy of 85.8% classified the rooms into three categories: small, large or hall. Using microphone arrays and analyzing the RIR obtained from them in [12] the authors are able to estimate the room geometry by exploring the properties of Euclidean distance matrices and show that under certain conditions the first order echoes provide a unique description of the geometry of convex polyhedral rooms. In [22] an indoor positioning device is presented using the RIR as the "fingerprint" to identify the room in which the system is located, the accuracy varies depending of the type of room, ranging from 65% in the identification of living rooms to 95% respect to bathrooms. The work in [44] shows a dataset of RIR in which people was standing in the rooms and relocated to different positions inside it to generate samples. The author shows the potential of using machine learning approaches over the dataset for tasks related to localizing, identifying and detect humans in those rooms.

2.6.2.2. Acoustic Object Shape Estimation

Acoustic perception of shapes are explored in [25], this work proposes the use of acoustic standing waves to recognize the size and shape of a target object analyzing the distance spectrum profile of the standing waves created by the interference between transmitted and reflected sounds, In this work Manabe et al. were capable of estimate the dimensions of a rectangular board of 1.2m x 0.9m and the mean absolute error of the estimation was 0.0194m, with this result they demonstrated that the spectrum profile encodes information of the target and can be use to estimate the shape of an object. The approach presented by Brunetto et al. in [5] result in the developed of the Batvision system capable of see deep spacial layouts by listening to echos recorded by two microphones as artificial ears. For Batvision a dataset is created using mobile robots that generate acoustic chirps signals at the same time a deep image is captured, with this dataset of pairs audio-image an adversarial discriminator is trained to generate deep images based on sound chirps. The system was capable of predict detailed indoor scenes and obstacles with a L1 loss as low as 0.072, sometimes even outperforming a deep mapping created by a stereo vision algorithm. Following the findings in Batvision the work presented in CatChatter paper [37] shows two separate results, first an improved version of Batvision is developed in which an input encoder from the model was removed, this encoder was processing the acoustic spectrogram before reaching the adversarial discriminator and was causing that spatial inputs from the spectrogram were lost at the time it reach to the U-Net decreasing the inference capacity of the model. The other result presented in [37] was from a new system developed that use lidar to create 3D, 360 degrees depth images together with an increase in the acoustic perception from 2 microphones to 4 in order to capture echoes from all the directions. The results obtained by this new method are similar to the ones reported in Batvision [5] but the field of prediction due to 360 deep information and the extra 2 microphones was 4 times bigger than the obtained with deep camera solution and two microphones system. To detect

basic shapes and deformation acoustic sensing was used in [30] by Rothstein et al. to identify and monitor small hollow object from inside. The deformation of the shapes are created by generating similar shapes with alter dimensions or by blocking a certain part of the object interior. A square wave sound signal is emitted in the interior of the objects and a 4 microphone array records the echos produced in the interior. Acoustic features are created from these signals and a classifier is trained to distinguish the shape and deformation of the object under study. Their work presented that for the objects the shape can be identify with 80% accuracy, the existence of deformation with 100% accuracy and localize the deformation with 90% accuracy.

Chapter 3

Hypothesis and Objectives

3.1. Hypothesis

The compendium of work presented in the Chapter 2 demonstrated that is possible detect with some precision the source of a sound in a tridimensional space, recognize a particular room by its unique acoustic and even create a deep image of segments of this room using only audible sound echoes and infer shapes of object based on acoustic features. Those results support the idea that for a given object inside a enclosed space is possible to learn a function that map the RIR to the location of the object in that space. Based on these ideas our hypothesis is:

"An algorithm pipeline with beamforming and ESN that utilize the rich spatial-geometrical information comprehended in a set of RIR of acoustic signals can infer the location and orientation of a cube of 20cm side allocated inside an small enclosed space of near of $1m^3$ [1], this with a movement distance resolution of at least 5cm and achieve

[1] Region with acoustic superposition found by [5]

less than 2cm of error in position and 10 degrees in orientation[2] with an error less than 6 degrees in the estimation from two consecutive positions."

3.2. Objectives

The current research established the following objectives:

3.2.1. Main Objective

To model and evaluatea processing pipeline using beamforming together with CNN and ESN to transform a multichannel RIR acoustic signals to spatial coordinates and angles that situate an object within an enclosed space.

3.2.2. Specific Objectives

1. Collect a dataset of acoustic responses of an object positioned in different locations inside a room.

2. Develop a reference method using CNN to learn from the dataset a function that estimate the coordinates and orientation of an object inside the space.

3. Develop a resources eficient method using ESN to learn from the dataset a function that estimate the coordinates and orientation of an object inside the space.

4. Build a beamforming stage that divide the measure space in four regions of acoustic sources.

5. Evaluate experimentally the accuracy of the estimators.

[2]Based on the size of the objects and maximin deformation presented in [30]

3.2.3. Deliverables

Specific Objective 1: A dataset will be delivered containing acoustic recordings of echoes in the form of time series and its corresponding object location-orientation.

Specific Objective 2: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with CNN.

Specific Objective 3: Thesis document with detailed description of the concepts, observation and results analysis that justify the system design with ESN.

Specific Objective 4: Thesis document with detailed description of the concepts, observation and results analysis that justify the beamforming.

Specific Objective 5: Thesis document with analysis and evaluation of the proposed solutions considering performance metrics under experimental operation.

3.3. Scope and Limitations

Scope and limitation 1: The development of the measure box and the collection of the dataset will be part of the work.

Scope and limitation 2: Only two ML approaches will be considered: CNN as the ground truth and ESN as a eficient solution.

Scope and limitation 3: The proposed system will not work on real time, even for the experimental evaluation the data will be first collected and later presented to the model in order to obtain the evaluation metrics.

Scope and limitation 3: The programming language will be python, the hardware to process the data will be a desktop personal computer.

Chapter 4

Methodology

The following chapter details the physical setup to develop the measurements that constitute the dataset under study, also present a description of testbench and processing pipeline in which the data flows from the acoustic inputs towards the desire location-orientation output.

4.1. Experiment Design

The process starts with the data collection under an experimental setup, the data will be curated to eliminate random noise and presented as a time series dataset. The resulting dataset will be used to train and test four neural network pipelines. The predicted location-orientation error is calculated and a multifactor Analysis of Variance (ANOVA) is proposed as the main criteria to evaluate the significant variance of the error of each processing pipeline used to estimate the location and orientation of an object in a room.

4.2. Experimental measurement configuration

The research require the development of a manipulable and fully controllable environment in which the acoustic measurements are going to be performed. A small box type room is the volumetric cubic space used as the experimental container. The container used for the experimental measurements is a cubic box with sides of 1m in length and a volume close to $1m^3$, the concept is show in Figure 4.1 (a), three of its six walls are built from porcelain tiles with a thickness of 1cm, one wall an the floor are made of solid concrete, while the remaining wall used for observation is made of acrylic with a thickness of 0.5cm. The clear acrylic sheet allows direct observation of the movement mechanism inside the box. Unlike the other walls, which are permanently bonded, this acrylic wall can be removed to provide access to the interior of the measurement box. The movement mechanism inside the box consists of a robotic arm mounted at the center of the floor. The arm has four Degrees Of Freedom (DOF) and is used to move and change the position and orientation of a small cube with 20cm side length attached to its end. The cube is made of plastic, with walls of 2mm thickness. For the measurements, a mono speaker is placed at the center of one wall to generate the acoustic stimulus inside the system. On the opposite wall, four capacitive microphones are positioned 10cm away from each corner, with the purpose of capturing the acoustic response of the measurement box. The implementation of the measurement box was build inside a storage room and is presented in the Figure 4.1 (b).

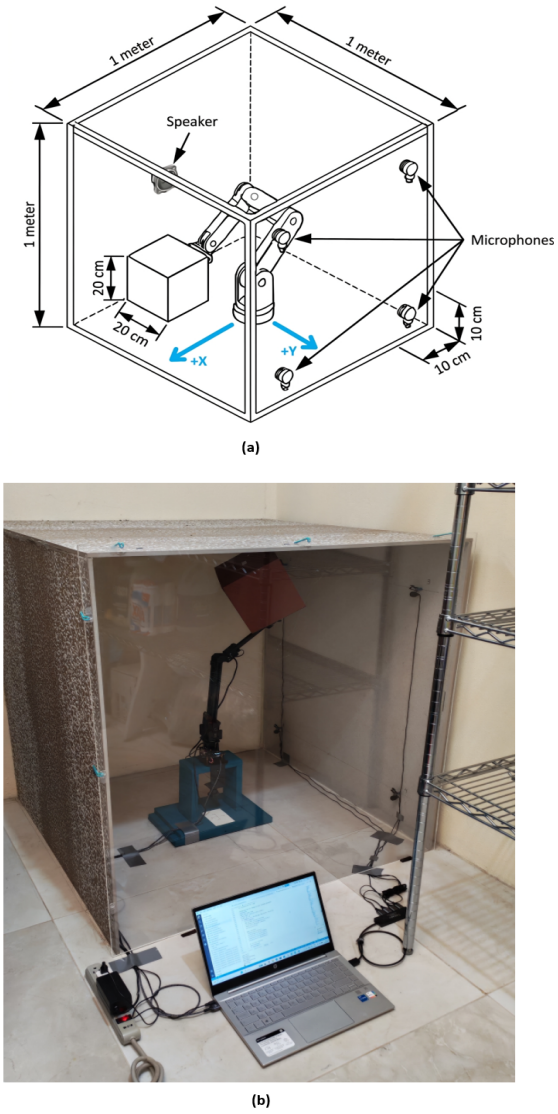


Figure 4.1: Measure box: (a) Conceptual sketch for the acoustic measurement box, and (b) actual implementation for the data acquisition task.

4.3. The Audio-Location Dataset for Model Training and Testing

The dataset was collected through twenty-six recording sessions. In each session, the robotic arm maintained a fixed angular value at its base and raised the cube to specific coordinates and orientations. The initial orientation of the cube was set with

its sides parallel to the measurement box walls. The variables modified during each measurement were the cube's location and orientation. For each location, nine different orientations were recorded. After completing these nine measurements, the robotic arm lowered the cube by decreasing its position along the Z axis, and the recording process was repeated for the new location. This procedure continued until the cube reached the bottom of the measurement box.

Audio from four channels was recorded. The acoustic signal was emitted four times, and the response was captured one microphone at a time. Each recording session consisted of 270 recordings, each containing four channels. A change of 10 degrees in the base of the robotic arm differentiated each recording session, producing a cylindrical envelope for the measurement area.

The coordinates of each measurement, obtained from the robotic arm, are shown in Fig. 4.2. The distribution of points is presented in the plot, each blue column in the three dimensions corresponds to the points in a recording session, and each point containts the nine orientations captured at that location. The total number of sessions was limited by the rotational constraint of the robotic arm's base, which allowed only 270 degrees of angular movement in its base. In total, 26 sessions were conducted, resulting in a dataset of 7020 recordings.

The source audio signal was obtained by arbitrarily extracting a 0.3-second segment from an acoustic melody. The spectrum of the signal has a bandwidth that reaches 10 kHz, with higher-magnitude components concentrated at low frequencies. Before reproducing the source signal, the recording process is initialized by opening the capture window for one second. This procedure is repeated four times, once for each microphone. After completing the four recordings, the cube is moved to a new orientation until it complete the nine possible and after this it moves to a new location.

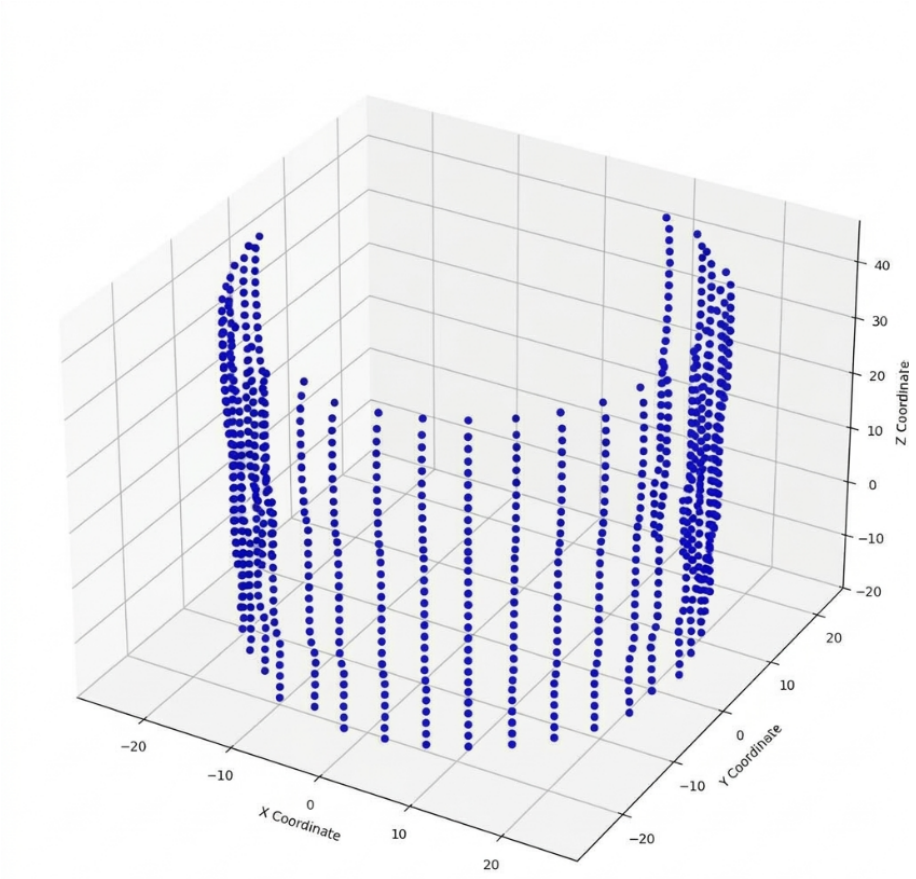


Figure 4.2: Scatter plot of cube locations for each recording in the dataset, the blue columns represent the 26 sessions, and each point in the column represents the nine orientations captured at that location.

4.4. The Audio-Location Processing Pipeline

In figure 4.3 can be observed the processing pipeline for the acoustics signals. From the measurement box, the microphone array records four signals to output four audio channels, these channels are used to feed the pipelines. First in AudAlign step the audio signals are aligned due to the temporal unpredictability in the recording start time caused by the OS process scheduler, the signals from the four channels are aligned by exploiting the fact that the distance from the source to each microphone is identical. Therefore, the magnitude of the direct sound can be used as an alignment reference for the signals. In the same step, a signal-integrity validation is performed in

which each recording is divided into its main body and tail (reverberation segment), from the main segment, the signal power is computed, and from the tail, its duration is extracted. Signals whose power and tail length fall within predefined limits are considered valid and included in the dataset. The resulting dataset is used in four diferent scenarios, two of them use a CNN-based pipeline and the other two use an ESN-based pipeline. The CNN-based require the time series data to be transformed into an image representation, while the ESN-based pipelines learn from an audio to spectral sequence series transformation. Additionally, a beamforming approach is suggested to address the superposition creating a segmented origin representation of the acoustic reverberation, dividing the room space into four different regions. The sound originating from these regions is contained in a group of newly created channels that will also feed both CNN and ESN estimators.

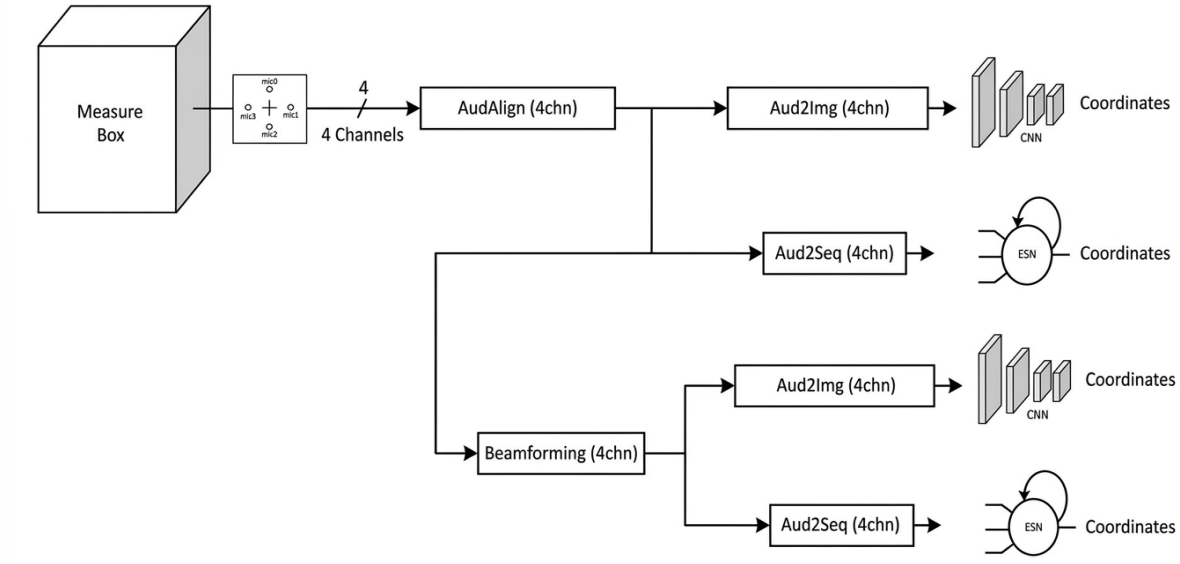


Figure 4.3: Acoustic to location processing pipeline

4.5. Factors and Levels

Factors are the independent variables that are going to be manipulated in the experiment, while levels are the different values that each factor can take. The factors and levels are selected based on the experimental design and the research questions being addressed. In this experiment, the factors include beamforming and network topology. Each factor has specific levels that will be tested to evaluate their impact in the task of estimating the location and orientation of the cube in the room.

For the experimental analysis, the factors and their respective levels to consider are summarized in the table 4.1. The network factor has two levels, The CNN is the reference model, the one that will be used as a baseline for comparison considering that once the audio signals are transformed into an image representation, the CNN is a well-established model for image processing tasks becoming a good candidate for the task of estimating the location and orientation of the cube in the room, the other level of the network factor is the ESN, which is an memory and training efficient model that will be evaluated against the CNN. Also a beamforming factor is considered, which has two levels, with and without beamforming. The combination of these factors and levels will allow for a comprehensive evaluation of the performance of the different processing pipelines in estimating the location and orientation of the cube in the room.

Table 4.1: Factors and levels for the experiment.

	Factors	
	Network	Beamforming
Levels	CNN ESN	With Without

4.6. Response Variables

The networks in this research are configured as regression models, the outputs are the predicted coordinates (X, Y, Z) and orientation of the cube in the room. The response variables are the error between the predicted and actual values of these outputs, which are calculated as mean of the absolute difference between the predicted and actual values as shows in equation 4.1. The Mean Absolute Error (MAE) is used as the performance metric for evaluating the models. "The MAE is a widely used metric for regression tasks, as it provides a clear measure of the average magnitude of errors in predictions, without considering their direction. Lower MAE values indicate better model performance, as they reflect smaller discrepancies between predicted and actual values".

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4.1}$$

4.7. Data Recollection for Model Evaluation

To evaluate the performance of the models, a five-fold cross-validation approach are used. In this approach, the dataset is divided into five subsets, and the model is trained and tested five times, each time using a different subset as the test set and the remaining subsets as the training set. This process allows for a more robust evaluation of the model's performance and helps to mitigate any potential bias that may arise from using a single train-test split. The total 7020 samples are divided into five folds, with each fold containing near to 1,404 samples (A 80:20 ratio). The performance of the model is evaluated considering the Mean Absolute Error (MAE). The structure of the five-fold cross-validation and the corresponding performance metrics are summarized in Table 4.2. Due to the factor and levels we have 4 model variation to consider, and for each model variation the coordinates X, Y, Z, and orientation are evaluated, resulting

in a total of 20 MAE to obtain for evaluation.

Fold Iteration	Training Set	Test Set	Performance Metric
Fold 1	Fold 2, Fold 3, Fold 4, Fold 5	Fold 1	MAE_1
Fold 2	Fold 1, Fold 3, Fold 4, Fold 5	Fold 2	MAE_2
Fold 3	Fold 1, Fold 2, Fold 4, Fold 5	Fold 3	MAE_3
Fold 4	Fold 1, Fold 2, Fold 3, Fold 5	Fold 4	MAE_4
Fold 5	Fold 1, Fold 2, Fold 3, Fold 4	Fold 5	MAE_5

Table 4.2: Five-fold cross-validation structure and performance summary.

4.8. Statistical Tool Used for hipotesis Testing

The groups of MAE obtained from the five-fold cross-validation for each model variation are considered as dependent samples because the same fold split of the dataset is used for each model evaluation. Two aproaches can be followed to evaluate the significance of the differences in the MAE between the different model variations. The first approach is to use a repeated measures ANOVA, which is a statistical test that compares means across multiple related groups. in order to use this approach, the assumptions of normality and sphericity must be met. If these assumptions are not met, a non-parametric alternative such as the Friedman test can be used. The Friedman test is a non-parametric test that compares the ranks of the data across multiple related groups. It does not assume normality and is suitable for small sample sizes. In the present work The Friedman test is used as the recomended option to determine if there are significant statistical differences in the MAE between machine learning models [29].

In the friedman test, Instead of comparing the means of the groups, the Friedman test ranks the data within each fold. The ranks are then summed for each model variation, and the test statistic is calculated based on these rank sums. Finally The resulting values are compared to a chi-square distribution to find the p-value, which indicates the probability of observing the data if the null hypothesis is true [18].

The hypotheses for the test are as follows: H_0: All models perform equally. H_1: At

least one model's performance is significantly different.

If the $p - value < 0.05$, reject H_0 and a post-hoc analysis is required to determine which specific models differ from each other.

Chapter 5

Design

"Another quote."

Another Author

This chapter presents the models use to test the hipotesis. As the ground truth a CNN estimator is selected to explore the the capability of a model to learn and estimate from the acoustic signals the location of an object in a tridimentional space. The input trasformation to images and the changes requiered to the red to work as an estimator are mentioned. Also, in this section the design of a memory eficient network in the form of an ESN estimator is presented. A justification for The ESN model architecture selection, the input trasformation and training and evaluation process is provided. Finally the Beamforming mechanism intended to compress information and highlight acoustic echoes from specific regions is described in detail.

5.1. CNN Estimator

The related work presented in seccion 2.6.2 provide insights respect the potential use of neural network to estimate the location of an object using acoustic echoes, and considering the especific findings in [30] is that a metric is extrapolated for the magnitude of estimation error. Due the fact that the dataset was new and unexplored, the model choose to establish a reference point was a CNN, this with the premise that a CNN have enogh representational capacity to learn from an especific acostic trasformation the function that maps the input features to espatial coordinates, providing valuable information if our original goals can be achieved with a more restricted network as is the case of an ESN.

5.1.1. Log-Mel spectrogram as images

To employ a CNN to train on the dataset a Log-Mel transformation is perform to the audio signals in order to obtain spectograms of images that can be use as input for the network. The audio signals have length of 35000 samples, and a Short-Time Fourier Transform (STFT) using a window length of 2048 that provide a good frequency

representation and a window step of 512 samples the frequency frames are obtained as detailed in equation 5.1 resulting in 68 frames. An aditional aditional mel filterbank is used to compress espectral data into 128 bands to help machine learning models to process the information more eficienty [1]

$$\text{Frequency frames} = \frac{signal_length}{window_step} \tag{5.1}$$

Finally to reduce feature variance in the mel bands a logaritmic trasformation is perform. The resulting log-mel matrix is transform to images of 570x370 pixes per location-orientation as the one show in Figure 5.1.

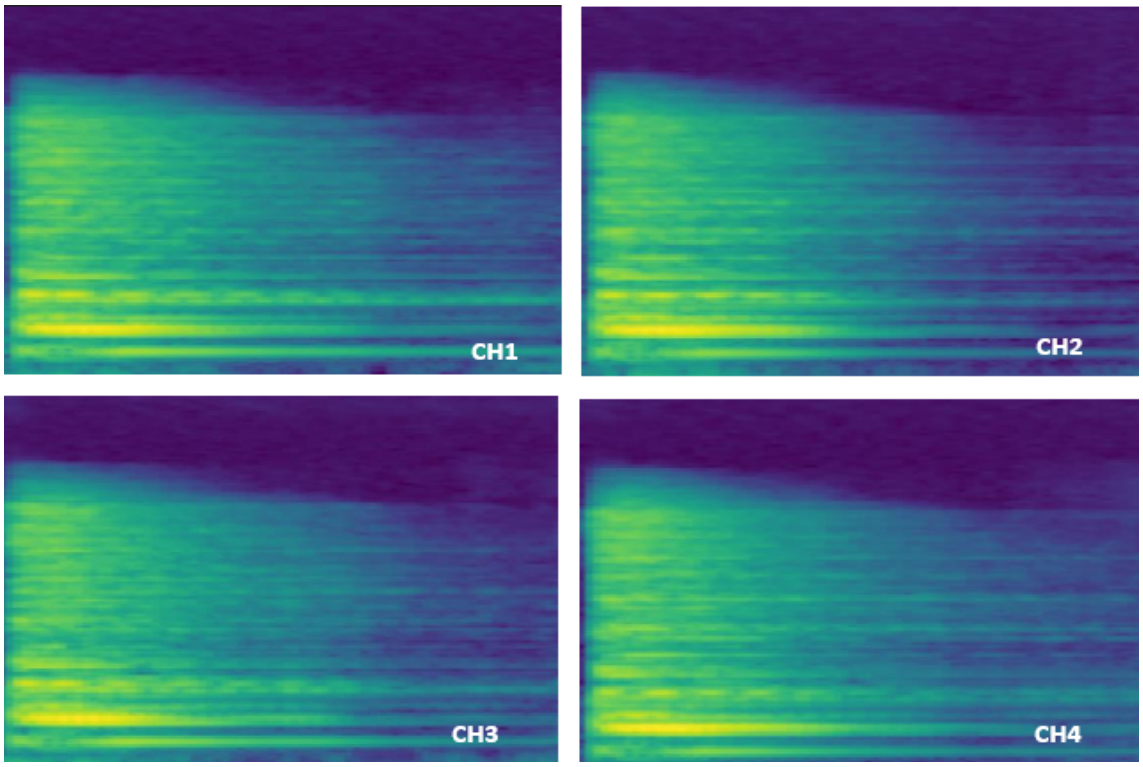


Figure 5.1: Log-Mel spectograms correspondig of the four channels of signals captured at session 1, location X=22.8, Y=-0.94 , Z=-9.43 and orientation 50 degrees.

5.1.2. CNN Architecture

The model architecture selected for the task of classifying and estimating the location–orientation of the cube inside the measure box was the ResNet-18 network, chosen for its robustness, simplicity, and suitability for inference in resource-constrained environments [3] (like the present in google colab notebooks). Two modifications were applied to the standard network structure. First, the input layer was extended to accept a stack of four RGB images, one corresponding to each audio channel. Second, for the regression task, the softmax output layer was replaced with a single output feature using a sigmoid activation function.

5.1.3. CNN Training and Evaluation

The network is trained to reduce the Mean Square Error for regression tasks, the optimization algorithm is the Adaptive Momentum Estimation. The training loop runs for 40 epochs, following the five fold paradigm, the training portion contains four folds and the test set is the remaining fold for a partitioning of 80:20 percent ratio. The epoch process happens in batches of size 32, for a total of 350 batches per epoch. The target values for X, Y and Z are normalized before the training process. A min-max normalization scales the value into the range $[0, 1]$ to fit the output response of the network. For X and Y an extra normalization step is performed since the path between these points describes a circular trajectory, the robotic arm's rotational movement at constant angular increments of 10 degrees causes the projections of these coordinates to be expressed as multiples of sine and cosine functions. Consequently, the distribution along the coordinate axes is not uniform and requires an arcsin transformation shown in equation 5.2 in order to result in a equidistant mapping.

$$label_arcnorm = \frac{\arcsin(2label_norm - 1) + \frac{\pi}{2}}{\pi} \tag{5.2}$$

After training the model is change to evaluation mode and the test set procesed to obtain the output from the estimator. A denormalization in the inverse funtion of equation 5.2 is perform and the results is use to calculate the estimation error as the diference with the ground truth coordinate value. The MAE, the error distribution and the percentage estimations per error range are calculated with the test results.

5.2. ESN Estimator

The ESN model is particularly efective learning from sequential data capable of learning from complex sequences of values due it rich temporal memory non linear dynamics, both atributes that share with the RNN but without the drawbacks of training the recurrent part of the network. The ESN tend to be a compact network, that the reason it is proposed in this document as a memory efficient model well suited to work as an estimator.

5.2.1. Mel spectrogram as sequence

For the ESN, the audio channels are cropped to only 20 000 sample to reduce the long tail with low signal power, the resulting signals are trasformend into sequence of Mel values, for this task, in similar way as te process described in seccion 5.1.1 a Short-Time Fourier Transform (STFT) using a window length of 1024 frames this time for speed up the transformation process and a window step of 512 samples was employ to obtain 40 frames. The Mel bank filter compress the data into 128 bands, a resulution adecuate for acoustic scene identification and direction of arrival identification [2]. The resulting bands are proceced by a decibel trasformation. The resulting spectrum sequence have a length of 40 values with 128 features per channel. A final step of feature scaling using min-max normalization is performed on the input features as a good practice to avoid problems with high magnitude variation between input values.

5.2.2. ESN Architecture

The development of the ESN architecture begins with the vanilla flavor of the ESN all set to work as an clasifier. for exploration and evaluation the network is first configure as an clasifier due that this task is easier than regression [40], so the earlier evaluations and changes on the ESN are conducted in it clasification mode.

The base ESN had as input four mel spectograms sequences of 128 bands each, for a total of 512 inputs, the neurons use tanh activation function for reservoir and the feedforward network. The output softmax layer clasifiy into 26 classes based on the session in which the audio is recorded.

The parameters for the networks are show in the table 5.1:

Table 5.1: Key parameters of an Echo State Network (ESN)

Parameter	Description
`reservoir size`	The total number of internal recurrent neurons within the reservoir, determining the dimensional capacity to represent complex dynamical states.
`spectral radius`	The largest absolute eigenvalue of the recurrent weight matrix, governing stability, memory decay rate, and the Echo State Property (ESP).
`sparsity`	The fraction of zero-weight connections within the reservoir matrix, decoupling internal sub-dynamics and lowering computation costs per update.
`leaky rate`	The leaking integration parameter $[0, 1]$, controlling how fast the reseirvoir forget previous states.

The spectral radius that control how memory decay in the reservoir is desire to have it as hier as posible in order that the network present a long term memory behaviour, the maximun value recommended had a boundary of 1 to ensure the ESP to warranty that the internal states of the reservoir will converge over time avoiding instability. For the ESN presented in this work a value of 0.99 was choose as spectral radius for the reservoir matrix. The leaky rate also controlls how fast the reservoir forget and it set to 0.1 to avoid memory loss and keep as much as posible of the history of the input sequence in the reservoir.

The sparity is keep to 0.1 for mantain a high conectivity of neurons in the reservoir. For the reservoir size, as the value set to this parameter is expensive in memory during training and is a direct factor determining the amount of trainable weights in the feedforward network, a search is perform to find a value with a good ration of acurracy/size. The figure 5.2 show the curve of the results obtained for the acuracy vs the reservoir size and can be observe that the acuracy increase as the reservoir size increase until reach the 1500 neurons, after this point increasing the size of the reservoir do not necesarily correspond to an improvement in the performance of the network.

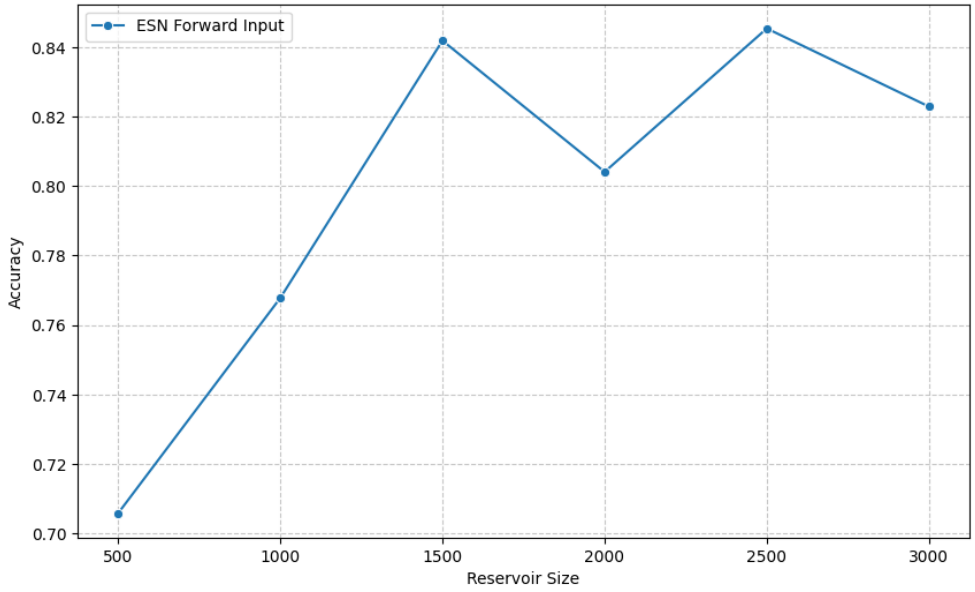


Figure 5.2: Network accuracy vs. reservoir size for 50 epochs runs.

A test with the described configuration shows acuracy below the 95% for training runs of 100 epochs, so a further improvement considered was to increase the features extracted from the reservoir using intermediate reservoir stages. Four configurations were considered: the original one, only capturing the state of the reservoir at the end, capturing a intermediate step at the midle of the sequence, three captures at 33%, 66% and at the end, and the las configuration with four captures at the 25%, 50%, 75%

and at the end of the sequence. The results are presented in table 5.2. with the higher mean value, a configuration of four capturing steps was choose as feature input to the feedforward network.

Table 5.2: Performance Across Intermediate Steps

Intermediate Steps	Run1	Run2	Run3	Mean
1 step	0.7692	0.7728	0.8262	0.7894
2 steps	0.8946	0.8291	0.8340	0.8525
3 steps	0.8547	0.8889	0.9024	0.8820
4 steps	0.9231	0.8996	0.9160	0.9129

The table 5.3 summarize the final values selected for the clasifier, with those values an acuracy of 98.42% was achieved on 150 epoch training. The results

Table 5.3: Echo State Network (ESN) Hyperparameters

Parameter	Value
Reservoir Size	1500
Spectral Radius	0.99
Sparsity	0.1
Leaky Rate	0.1
Intermediate Captures	4

The results with the clasifier demostrated that the ESN had the capability of learn from the sequence the characteristic patterns to distinguish the session pilar in which they were recorded. This network was the base for the regresion network. The output layer was replaced from 26 to 1, and two aditional hiperparameter for the network as it regresion form were defined using grid search: 1) The alredy defined sparcity value, and 2) To improve the learning capacity of the ESN, the single layer feedforward network is replaced by a two layer network with a hidden layer of magnitude defined as hidden size. The search results for these two parameters are presented in figure 5.3. Considering as criteria to choose the parameters a good performance of the network at the minimal size, the sparcity of 0.7 and a hidden layer of 20 is selected for the ESN estimator.

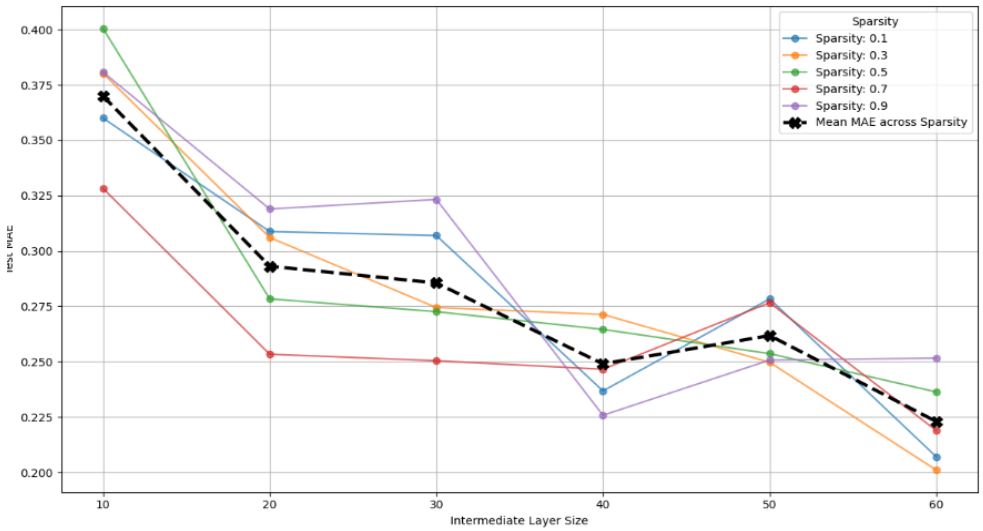


Figure 5.3: Test MAE as a function of Intermediate Hidden Layer Size for different Sparsity values

Table 5.4: Error Distribution Comparison Between Standard ESN and MultiHead ESN

Error Range (cm)	ESN Count (%)	MultiHead ESN Count (%)
$(-1, 1)$	39.80	76.65
$(-2, 2)$	62.31	95.01
$(-3, 3)$	76.72	98.38
$(-4, 4)$	85.16	99.09

The ESN estimator after training 1200 epochs produced a MAE of 2cm with the percentage sample per error range shows in table 5.4, in order to further reduce the MAE of the ESN, it architecture was enhanced to use multihead feedforward network. The reason for this change comes from the observation that the ESN perform well as a clasifier, been capable to distinguish recording session near to each other, once the fedforward network was configured for the regresion task the errors in estimation were even larger that the distance betwen adyacent sessions, suggesting the possibility that the network was not able to learn the mapping function from the reservoir features to the output coordinates. A key diference between the clasification and regresion tasks is that the clasifier had an independent output for each class, but not only the output, it had an entire feedforward network in the form of a single layer perceptron for each

class, so the network was able to learn a different mapping function for each class. The multihead feedforward network is a similar approach, but instead of having a single layer perceptron for each class, the learning task was dividen into three independent task, one network was dedicated to map the function for the positive coordinates, another for the negative coordinates and a third to predict the sign of the coordinate. The sign prediction network is resposable to select which of the other two networks will be used to predict the value of the coordinate. that provided a sustancial improvement in the performance of the ESN estimator. but still some small amount of estimation presented a large error in the case of misprediction of the sign of the coordinate, for this reason a forth network was proposed to estimate the value of the coordinate for the entire range of values, the output acuracy of this network is lower than the other arrangement, but is intended to be used as guidence for the other network, if the diference from estimation fromthe two networks is larger than a threshold, the network consider that a error inthe sign prediction network has ocurred andthe output is selected fromthe network of opposite sign. The architecture ofthe multihead feedforward network is shown in figure 5.4 and the results of the ESN estimator with this configuration is shown in table 5.4 with a MAE of 0.7cm and a significant improvement in the percentage of samples per error range.

5.2.3. ESN Training and Evaluation

Each feedforward head was trained independently, the learning rate was set to $5e-5$ for all the sub-network heads, the batch size was set to 32. The training was perform separately for each head. The sub-heads for positive and negative coordinates were trained using only the samples with the corresponding sign, while the overall coordinate prediction head and the sign prediction head were trained using all the samples, the sub-heads for positive, negative and overall coordinate estimation were trained using Adam optimizationfor 400 epochs, while the sign prediction head was

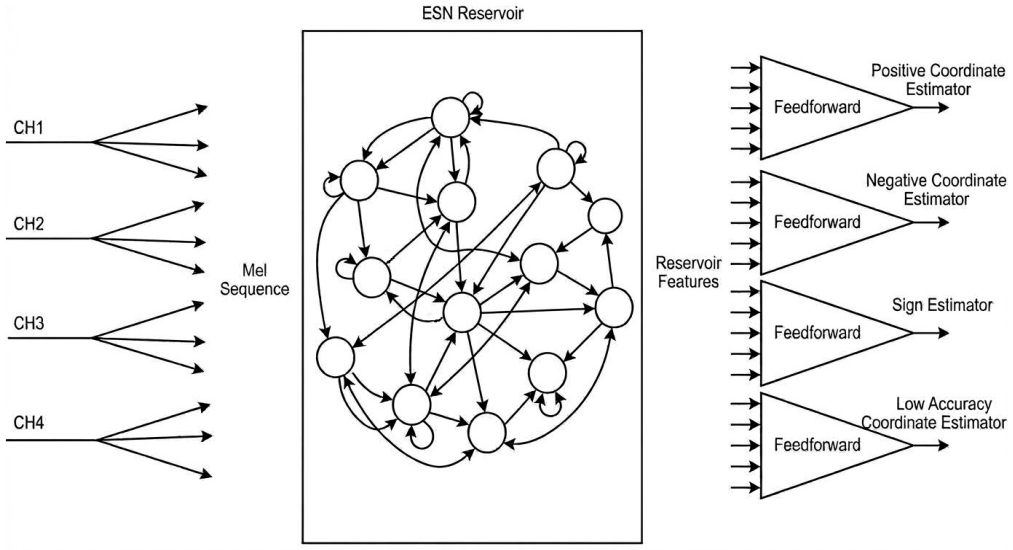


Figure 5.4: Multihead Network architecture for the ESN Estimator.

trained using Binary Cross entropy with Logits Loss for 800 epochs. The training process was performed using a fivefold cross-validation approach, with four folds used for training and one fold reserved for testing, following an 80:20 percent ratio. The target values for X, Y and Z are normalized as described in section 5.1.3, and a denormalization step is applied to the output of the network to obtain the final coordinate estimations. The evaluation of the ESN consider the results of the four sub-heads using a hibrid aproach, the output selection rather than relying solely on a single prediction head, the final coordinate is computed through a decision tree. The sign head predicts the sign of the target Conditional Path Selection:If the predicted sign is positive, the positive head output is chosen.If negative, the negative head output is chosen. This yields a candidate coordinate estimation, referred to as the Conditional Path prediction.An outlier mitigation is also performed, if the absolute difference between the Overall Estimator prediction and the chosen Conditional Path prediction is calculated.If this difference exceeds a threshold (100 cm), indicating a potential critical sign classification failure, the model falls back to the alternate sign's

estimator output to prevent high localization errors. the Algorith for the hibrid desition logic is shown in Algorith 1.

Algorithm 1 Hybrid Decision Logic for Coordinate Estimation with Outlier Mitigation

Require: Feature vector $\mathbf{x}$, Sign Head $\hat{s}(\cdot)$, Positive Head $\hat{y}_+(\cdot)$, Negative Head $\hat{y}_-(\cdot)$, Overall Estimator $\hat{y}_{\text{overall}}(\cdot)$, Outlier Threshold $\tau = 100$ cm
Ensure: Final Coordinate Estimation $\hat{y}$

// Phase 1: Sign Classification
pred_sign $\leftarrow \hat{s}(\mathbf{x})$ ▷ Predicts whether target is positive or negative
// Phase 2: Conditional Path Selection
if pred_sign ≥ 0 **then** ▷ Positive condition branch
    $\hat{y}_{\text{cand}} \leftarrow \hat{y}_+(\mathbf{x})$ ▷ Candidate conditional path prediction
    $\hat{y}_{\text{alt}} \leftarrow \hat{y}_-(\mathbf{x})$ ▷ Alternate path for fallback
else ▷ Negative condition branch
    $\hat{y}_{\text{cand}} \leftarrow \hat{y}_-(\mathbf{x})$
    $\hat{y}_{\text{alt}} \leftarrow \hat{y}_+(\mathbf{x})$
// Phase 3: Outlier Mitigation and Fallback Logic
$\hat{y}_{\text{ref}} \leftarrow \hat{y}_{\text{overall}}(\mathbf{x})$ ▷ Evaluate baseline overall estimator
$\Delta \leftarrow |\hat{y}_{\text{ref}} - \hat{y}_{\text{cand}}|$ ▷ Compute absolute discrepancy
if $\Delta > \tau$ **then** ▷ Potential critical sign classification failure detected
    $\hat{y} \leftarrow \hat{y}_{\text{alt}}$ ▷ Fallback to alternate estimator output
else
    $\hat{y} \leftarrow \hat{y}_{\text{cand}}$ ▷ Retain candidate conditional path prediction
return $\hat{y}$

5.3. Beamforming Implementation

5.4. Beamforming

5.4.1. Frequency Selection

In order to implement beamforming using the audio channels, the principal spectral components of the source signal used as the stimulus within the measurement box must be analyzed. For this purpose, the Python "XXX" libraries are used, and the frequency response of the audio signal is plotted. The frequency response plot is shown in Figure 5.5, while the top 10 frequencies with the highest magnitude are presented in Table 5.5,

where it can be observed that frequencies near 400 Hz exhibit the largest magnitudes. Based on this, 400 Hz is selected as the beamforming frequency.

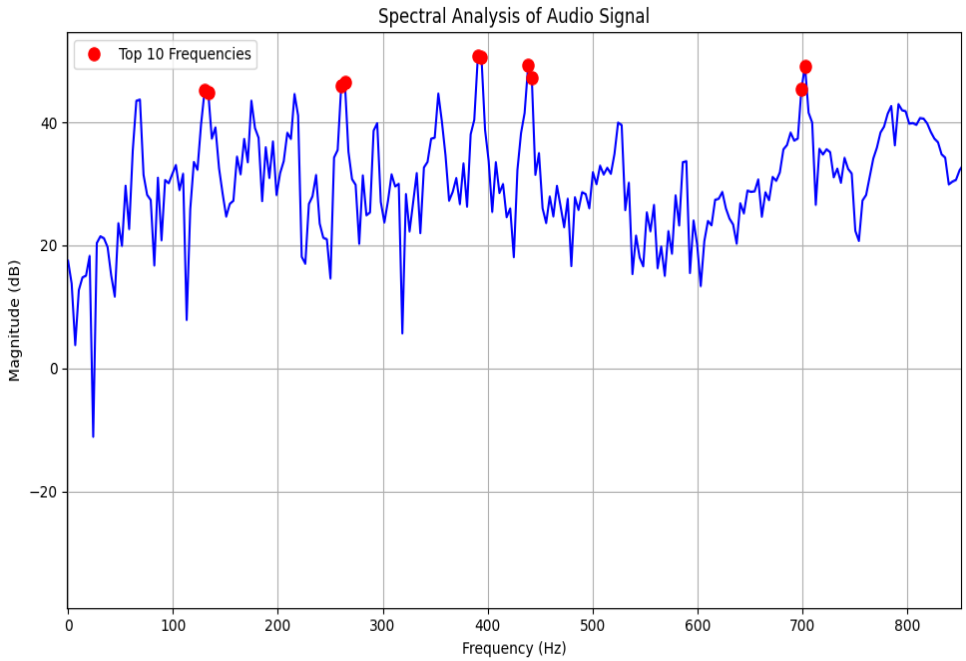


Figure 5.5: Frequency response of the stimulus signal showing the top 10 spectral components with the highest magnitude.

Rank	Frequency (Hz)	Magnitude (dB)
1	390.72	50.71
2	394.14	50.58
3	438.70	49.31
4	702.61	49.17
5	442.13	47.26
6	263.91	46.60
7	260.48	45.96
8	699.18	45.32
9	130.24	45.19
10	133.67	44.88

Table 5.5: Top 10 frequencies with the highest magnitude (dB).

Given that the microphones are placed in a plane parallel to the X-Z plane, forming pairs that enable binaural audition, measurements along the X and Z axes can serve as candidate inputs to evaluate the impact of beamforming on estimating

the cube's position. A geometric analysis of the measurement box is required. In Figure 5.6, the midpoint of the sides perpendicular to the X-axis is defined as a location of interest. These points will be considered the acoustic source for the beamforming procedure. Assuming the sound originates from this point or surrounding regions, it is necessary to compute the path-length difference traveled by the same acoustic wave arriving at both microphones. Using trigonometry, the distance from point E to G is determined to be 0.522 m, and the distance from E to H is 0.9862 m. Their difference yields 0.4649 m, which represents the extra distance the acoustic wave must travel to reach point H compared to point G. Assuming the speed of sound is 346 m/s—the approximate speed at a temperature of 26 °C [42], which closely matches the average temperature at the testing site—the time difference of arrival (TDOA) between both microphones is 1.34 ms, as expressed in Equation 5.3.

$$\Delta t_{\text{arrival}} = \frac{0.4642\,\text{m}}{346\,\text{m/s}} = 1.34\,\text{ms} \tag{5.3}$$

Selecting the highest-magnitude components around 400 Hz for beamforming, combined with the computed time difference of arrival, enables an analysis of how the signal's main spectral components behave over the travel time Δt. At 400 Hz, the signal period is 2.5 ms. For a propagation delay $\Delta t = 1.35$ ms (as illustrated in Figure 5.7), the acoustic wave arrives at point G in an inverted-phase condition relative to point H, causing destructive cancellation when the two signals are summed. Consequently, introducing a delay equal to half the period (1.25 ms) produces constructive interference for signals originating from point E at 400 Hz. Furthermore, as the reverberant source shifts away from this target location, destructive attenuation penalizes the gain at these frequencies, fulfilling the objective of the beamforming operation.

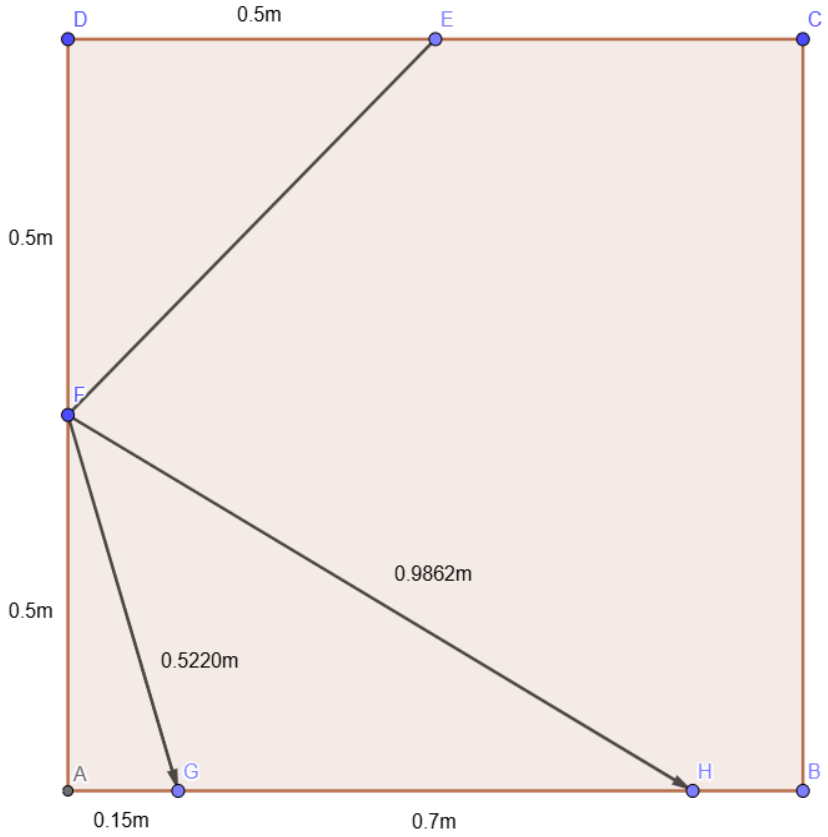


Figure 5.6: Geometric layout of the measurement box illustrating the acoustic path from the speaker located at E to the microphones positioned at G and H.

5.4.2. Audio compactation

5.4.3. Limitations and Requirements

A brief description of the design mechanism's limitations and requirements in accordance with the definitions in Scope and Limitations.

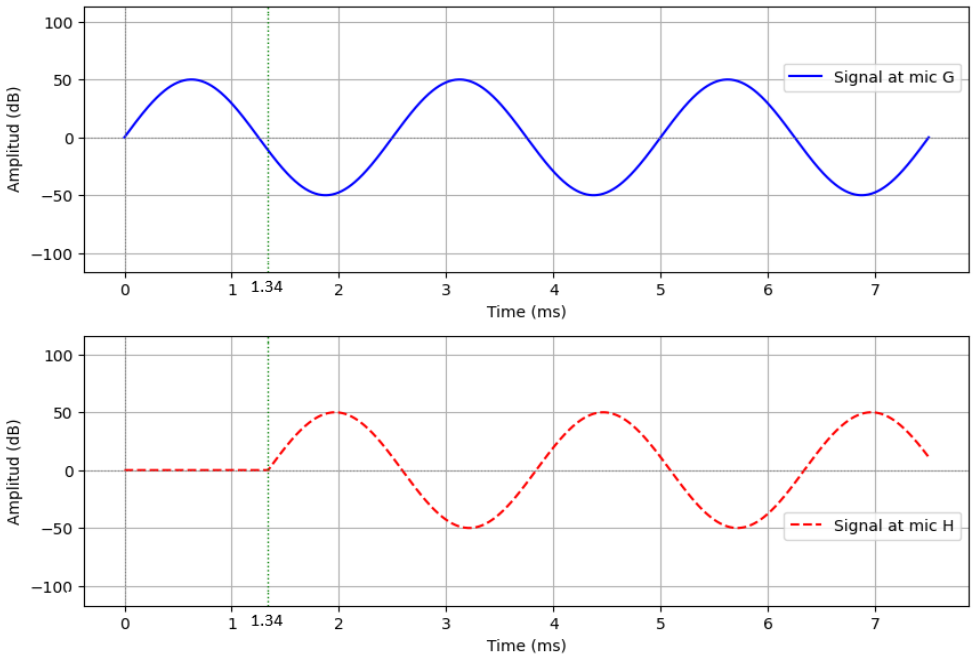


Figure 5.7: Acoustic waveforms at 400 Hz captured by the microphones located at points G and H.

Chapter 6

Results

"Another quote."

Another author

This chapter presents the results obtained while evaluating the proposed algorithm/mechanisms. The discussion and analysis of results are presented in the Discussion chapter.

6.1. CNN for Acoustic Classification

6.1.1. CNN Acoustic to Spatial Location Classification

The first evaluation of the CNN was on a classification task where it had to classify the acoustic location of the sound source from the 26 possible locations, each one corresponding to a specific recording session characterized by the same X, and Y location and only changing the Z location. Trained on 20 epochs, the network shows a accuracy of 99.4% on training and 98.59% on the test set. The confusion matrix is shown in Figure 6.1.

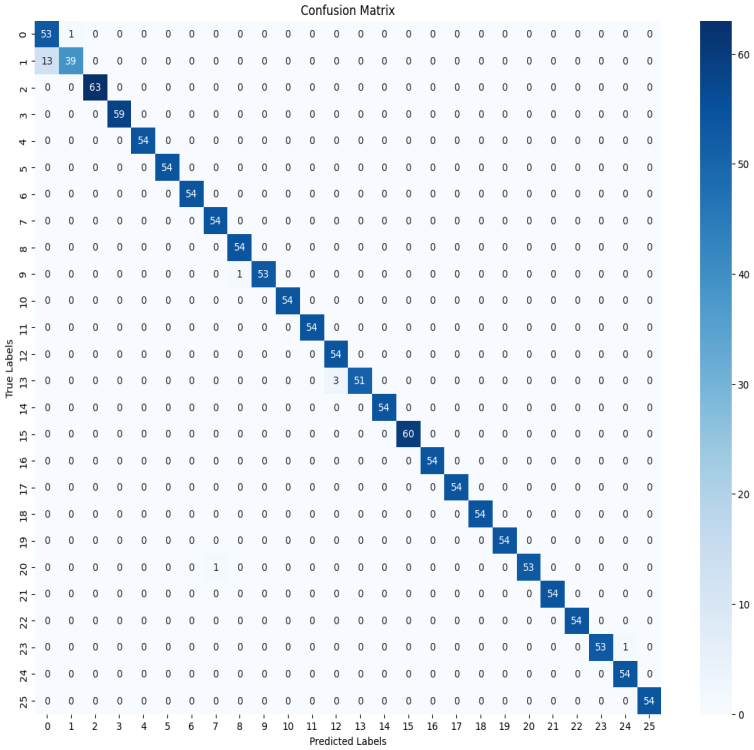


Figure 6.1: Confusion matrix for CNN acoustic location classification on test set.

6.1.2. CNN Acoustic to Orientation Classification

The second evaluation of the CNN was on a classification task where it had to classify based on the orientation of the cube from a total of 9 possible orientations. For each orientation the cube is rotated 10 degrees with respect to one of its centroidal axes. Trained on 20 epochs, the network shows a accuracy of 99.0% on training and 29.06% on the test set. The confusion matrix is shown in Figure 6.2.

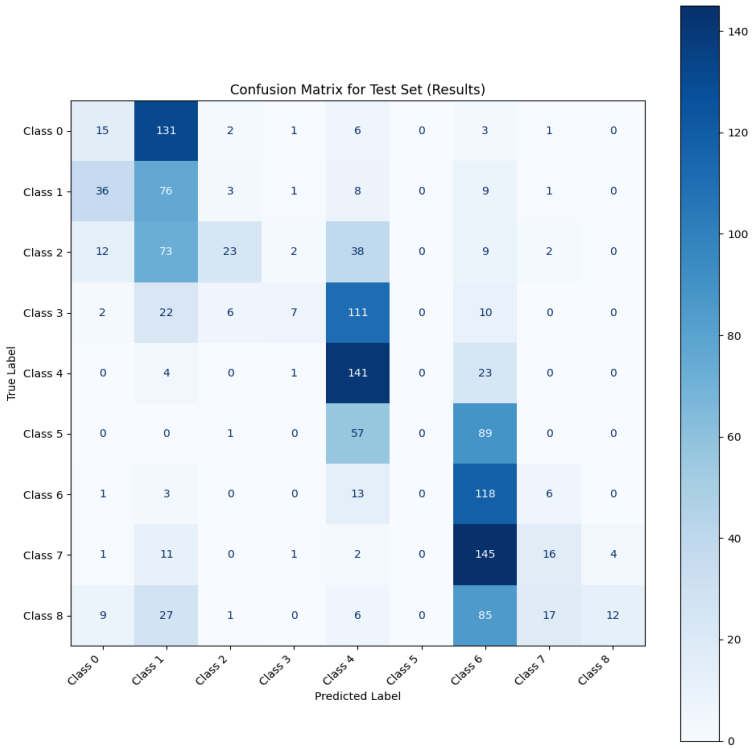


Figure 6.2: Confusion matrix for CNN acoustic orientation classification on test set.

A further exploration is carried out to analyze the performance of the CNN on the orientation classification task. As shown in Figure 6.4, three regions are defined to reduce changes in orientation for the other Euler rotational axes. For each of the three regions, the accuracy results are presented in Table 6.1 and the corresponding confusion matrix are presented in Figure 6.3.

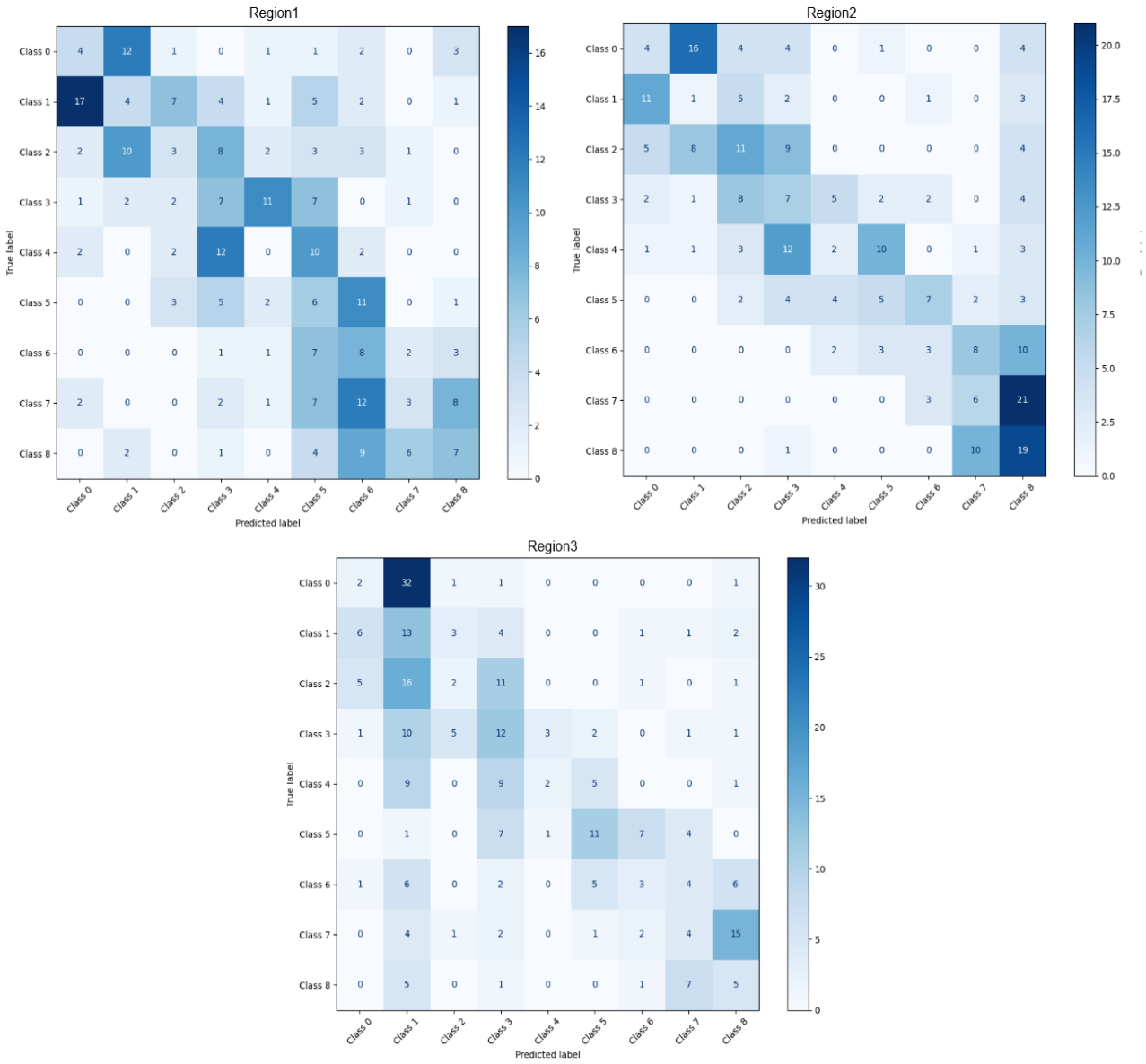


Figure 6.3: Confusion matrix for the three regions defined for orientation classification.

Table 6.1: Model Performance Across The Three Regions

Region	Training Accuracy (%)	Test Accuracy (%)
Region1	100.0	15.56
Region2	99.9	21.48
Region3	100.0	20.00

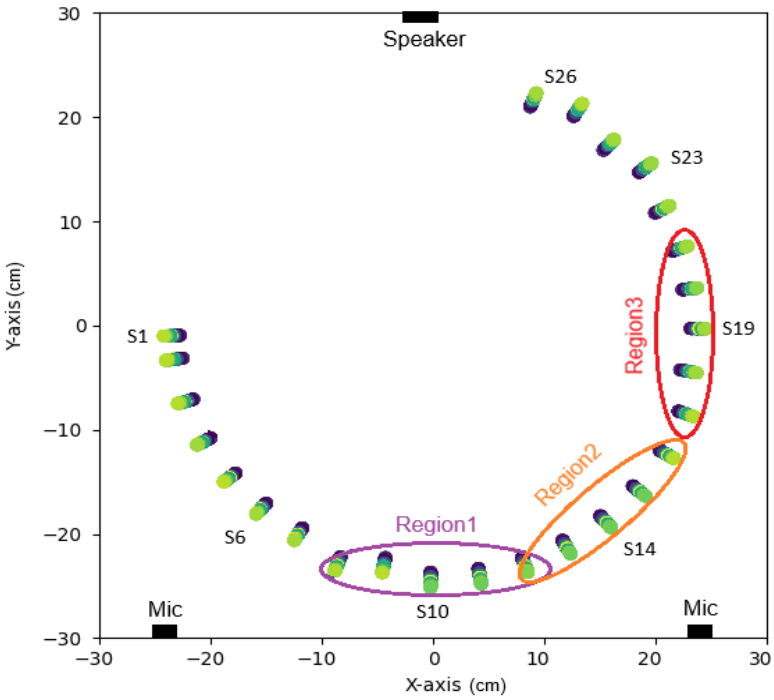


Figure 6.4: Session point used for cross-correlation analysis. the image also shows the three regions for analysis of the orientation classification task.

Also to investigate the similarity of the acoustic signal when the cube is rotated or the cube is moved, a cross-correlation analysis was performed. The selected sessions for the analisis correspond to a sample that comprends diferent position that goes from the nearest to the mics to the farthest in the area near the speaker. The results of the cross-correlation analysis are presented in figure 6.5.

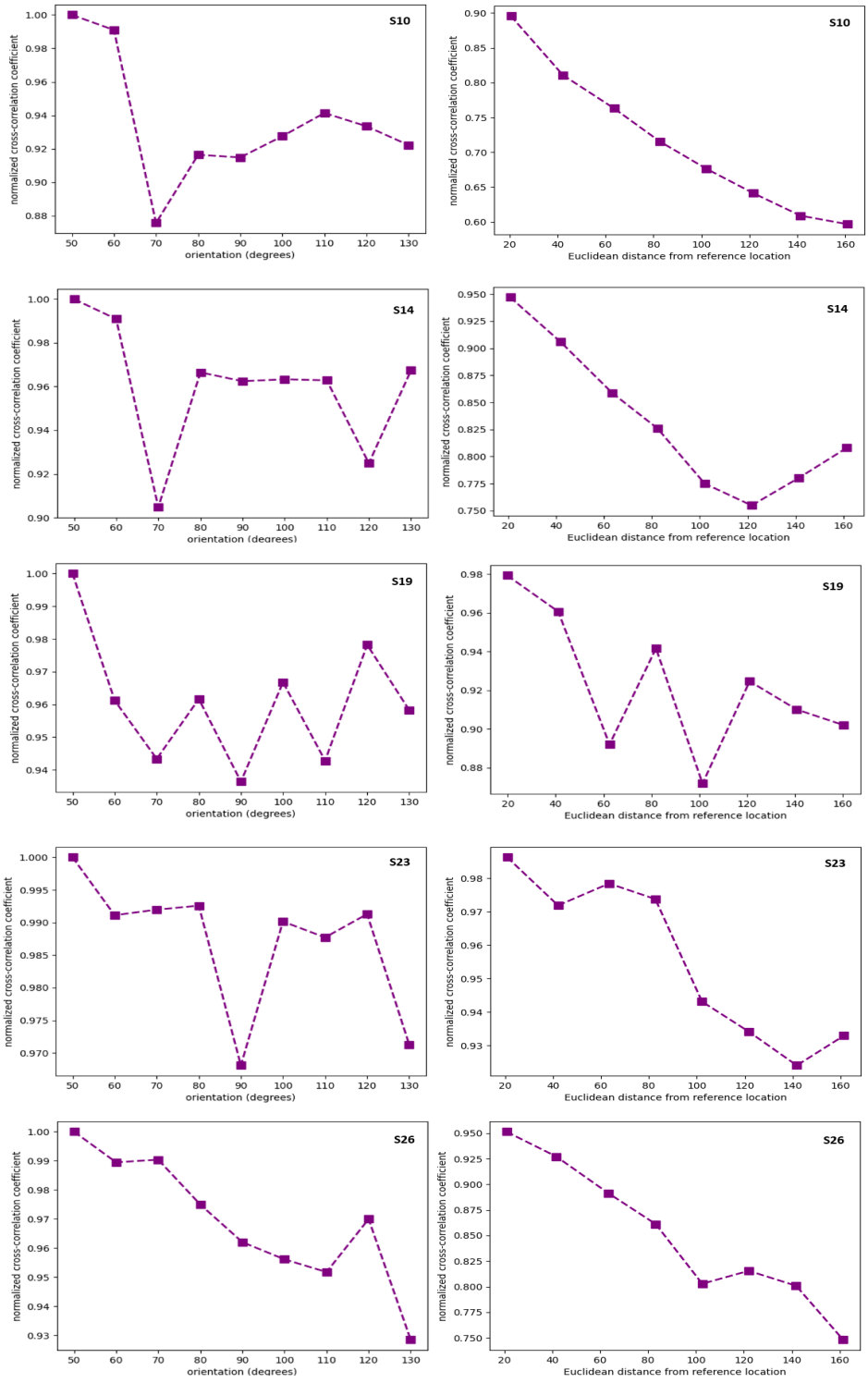


Figure 6.5: Cross-correlation results for distance and orientation for similarity analysis of the acoustic signal.

6.2. CNN Acoustic to Spatial Location Regression

The CNN was evaluated as a regression model to estimate the X, Y, and Z coordinates of the cube inside the measurement box. In all cases, it was trained for 40 epochs following a five-fold cross-validation scheme. The histograms of the estimation errors for the three coordinates are shown in Figures 6.6, 6.7, and 6.8.

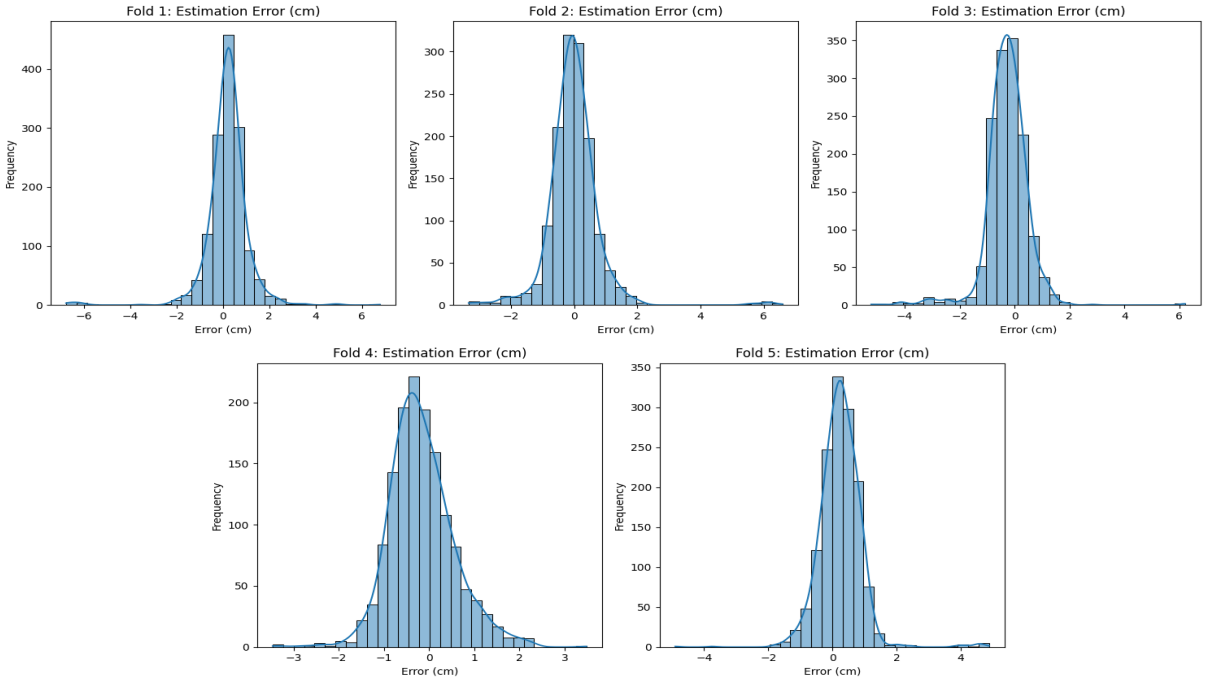


Figure 6.6: Histograms of estimation errors with CNN for X values across five folds.

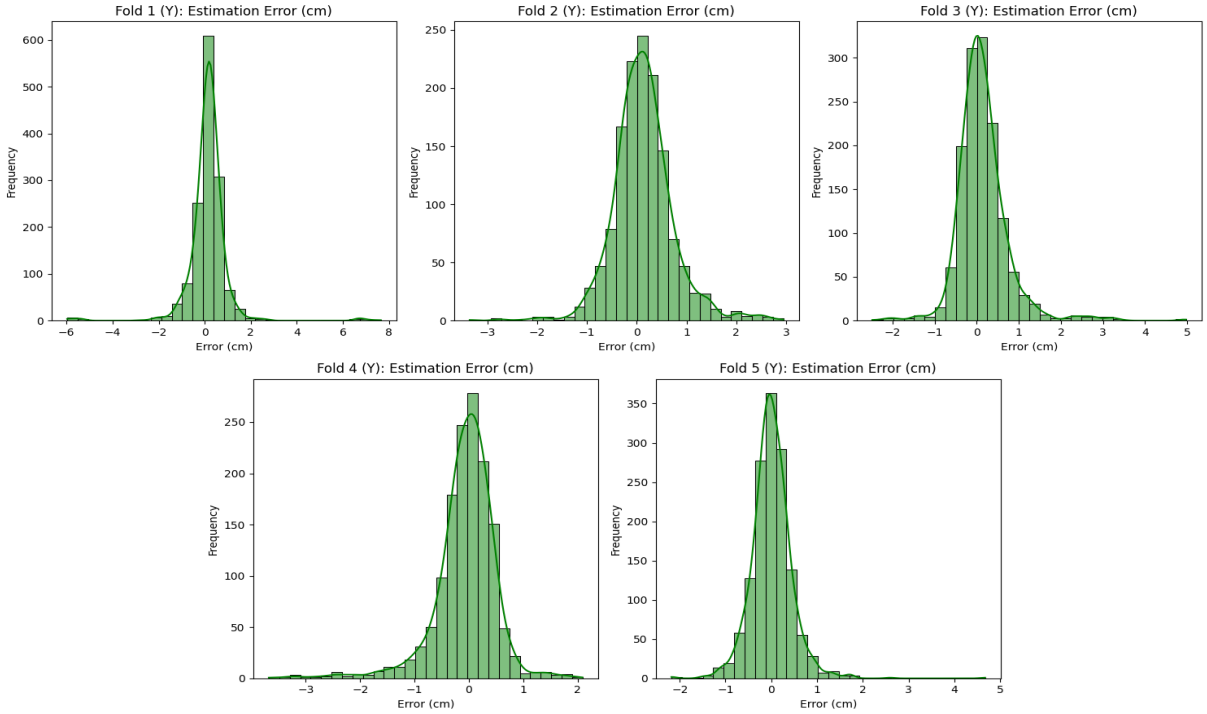


Figure 6.7: Histograms of estimation errors with CNN for Y values across five folds.

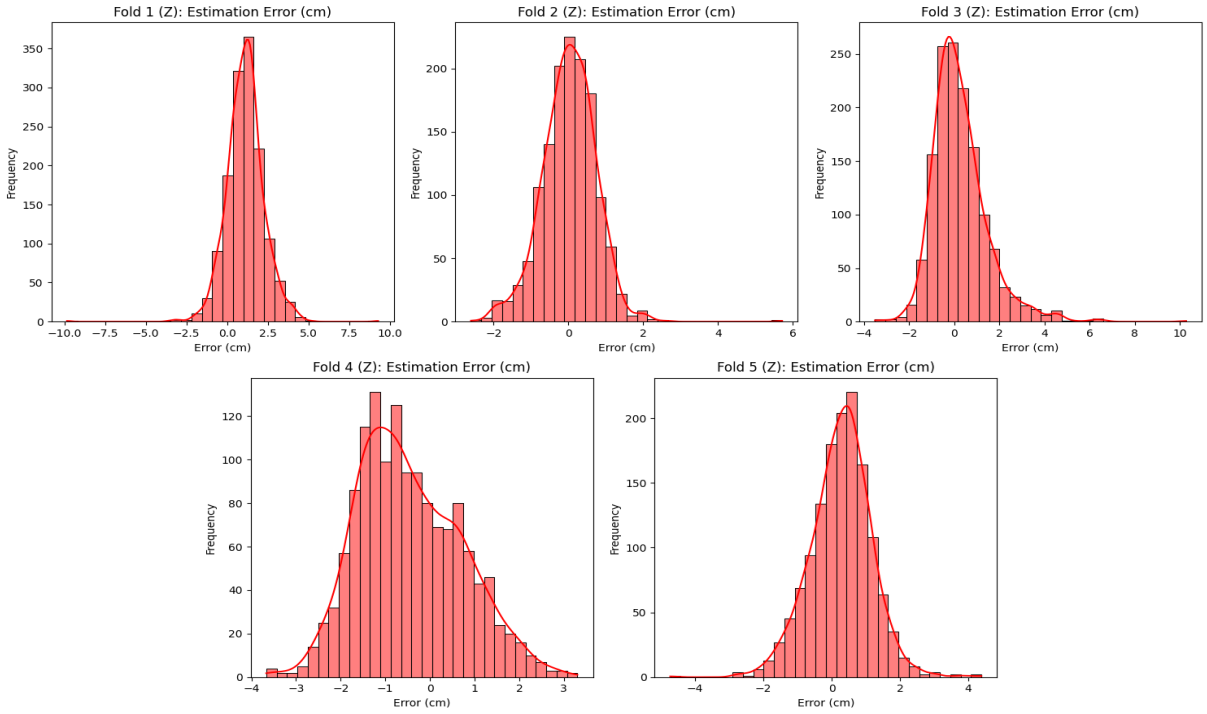


Figure 6.8: Histograms of estimation errors with CNN for Z values across five folds.

The main figure of merit for the regression model is the mean absolute error (MAE), which measures the average magnitude of the prediction errors regardless of their direction. The MAE values for the three coordinates across the five folds are presented in Table 6.2.

Table 6.2: Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	0.6019	0.5029	1.3013
Fold 2	0.5231	0.4467	0.5549
Fold 3	0.5569	0.4069	0.8829
Fold 4	0.5764	0.3836	1.0473
Fold 5	0.5062	0.3144	0.7316
Mean ± SD	**0.5529 ± 0.0384**	**0.4109 ± 0.0706**	**0.9036 ± 0.2847**

The second figure of merit is the Pearson correlation coefficient, which measures the strength and direction of the linear relationship between two continuous variables. It indicates how closely the data points cluster around a straight line. To understand how well the predictions of the regression model correlate with the ground truth, the Pearson correlation coefficients for each coordinate are shown in Table 6.10. In the evaluation of predictions against ground truth, a coefficient greater than 0.99 indicates that the model predictions are almost perfectly aligned with the actual targets.

Table 6.3: Pearson Correlation Coefficients between Ground Truths and Predictions

Folds	X	Y	Z
Fold 1	0.9970	0.9961	0.9979
Fold 2	0.9980	0.9985	0.9991
Fold 3	0.9969	0.9978	0.9982
Fold 4	0.9985	0.9977	0.9986
Fold 5	0.9978	0.9987	0.9987
Mean ± SD	**0.9976 ± 0.0007**	**0.9978 ± 0.0010**	**0.9985 ± 0.0005**

Another important metric for evaluating the regression model is the percentage of samples whose estimation error falls within a specific range. The percentage of samples

per error range for X, Y, and Z is shown in Table 6.4.

Table 6.4: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates

Axis	Error Range (cm)	Fold1(%)	Fold2(%)	Fold3(%)	Fold4(%)	Fold5(%)	Mean(%)
X	$(-1, 1)$	84.95	88.91	88.87	85.22	90.51	87.69
	$(-2, 2)$	96.77	97.45	97.45	98.73	98.79	97.83
	$(-3, 3)$	98.66	99.05	98.58	99.79	99.14	99.05
	$(-4, 4)$	99.02	99.34	99.36	100.00	99.36	99.42
Y	$(-1, 1)$	90.58	90.88	92.84	93.56	96.65	92.90
	$(-2, 2)$	97.47	98.03	97.94	98.44	99.71	98.32
	$(-3, 3)$	98.66	99.93	99.43	99.65	99.93	99.52
	$(-4, 4)$	98.66	100.00	99.86	100.00	99.93	99.69
Z	$(-1, 1)$	42.69	85.56	69.43	51.41	73.11	64.44
	$(-2, 2)$	81.58	98.98	91.35	91.09	96.93	91.98
	$(-3, 3)$	94.59	99.93	96.52	99.29	99.50	97.97
	$(-4, 4)$	98.95	99.93	98.58	100.00	99.79	99.45

The estimation error for each point in the dataset is presented in Figure 6.9. The graph considers the five folds, each evaluated while it served as the test set. The error is calculated as the absolute difference between the predicted and true values of the coordinate in the respective dimension, and is expressed in centimeters. Figure 6.10 presents the error of each coordinate with respect to its own axis, showing the X error as a function of X, the Y error as a function of Y, and the Z error as a function of Z.

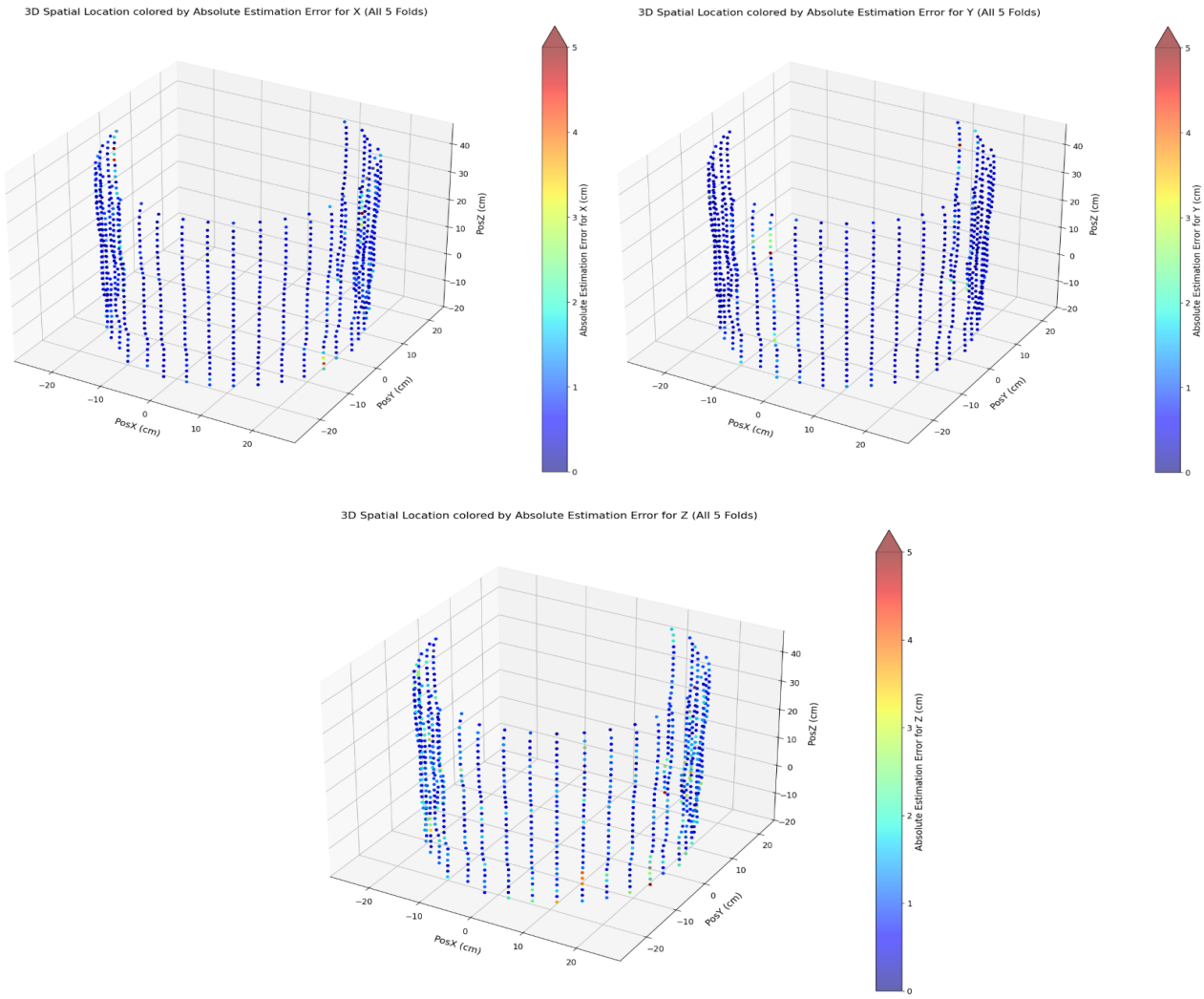


Figure 6.9: 3D scatered plot showing the absolute error value for all the points estimated in the five folds as test set.

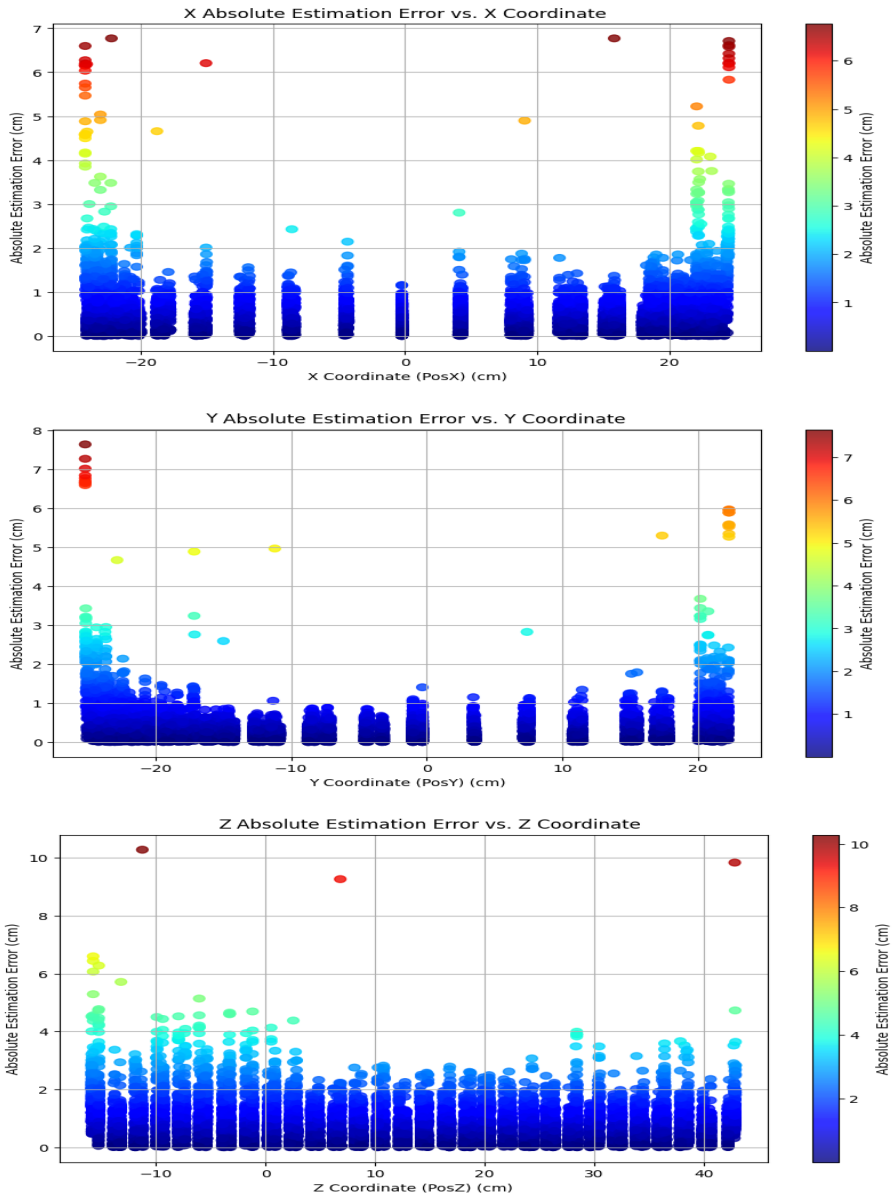


Figure 6.10: Scatered plot showing the absolute estimation error vs it respective coordinate axis.

6.3. ESN for Acoustic Classification

Previously described in section 5.2.2, the ESN first was evaluated as a classifier for the architecture definition and hyperparameters search. With the values shown in table 5.3 after training for 150 epochs, the ESN accuracy was 98.42%. The resulting

confusion matrix is presented in Figure 6.11.

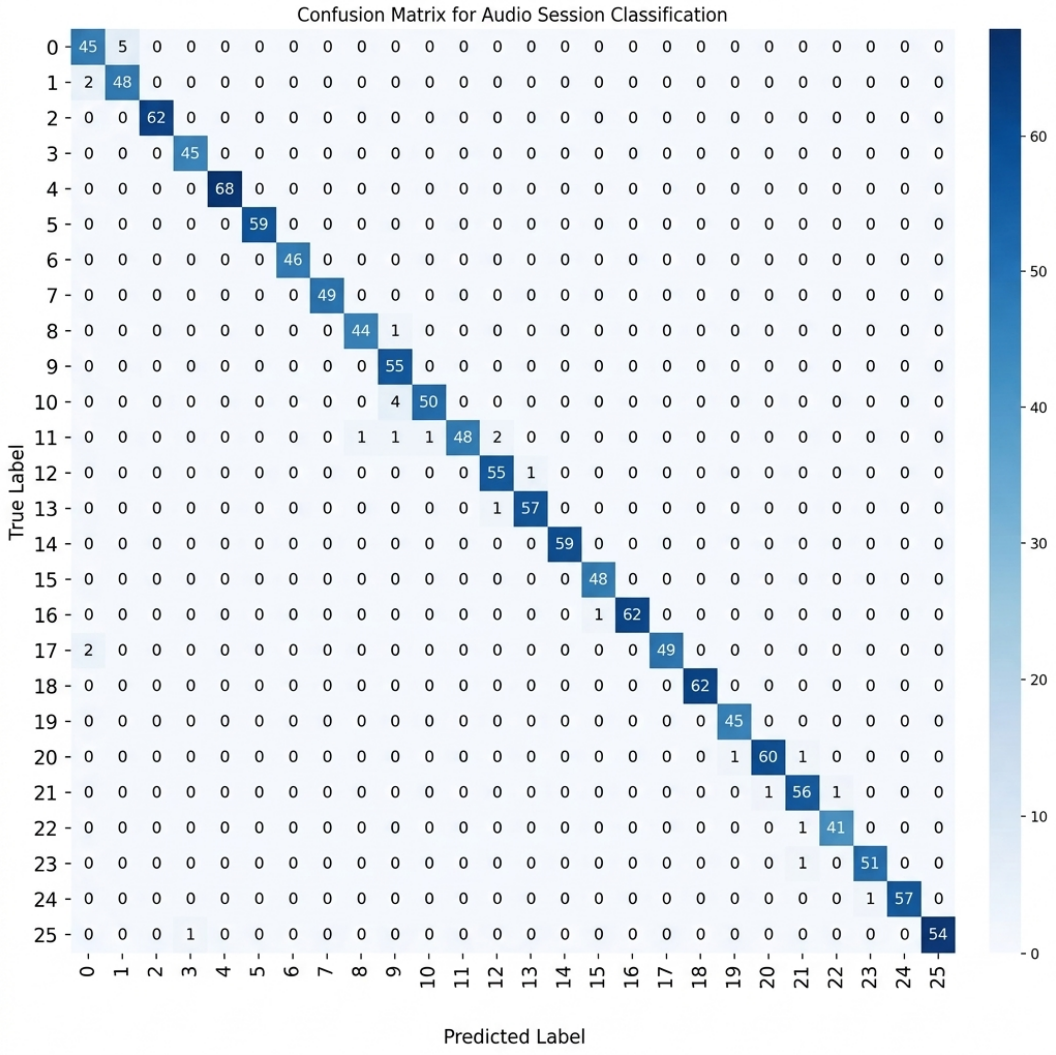


Figure 6.11: Confusion matrix for the ESN as result of the audio session classification.

6.4. ESN as regression model

Similar to the CNN for the ESN the histograms for the error distribution are presented in figure 6.12 figure 6.13 and figure 6.14, for X Y and Z coordinates respectively.

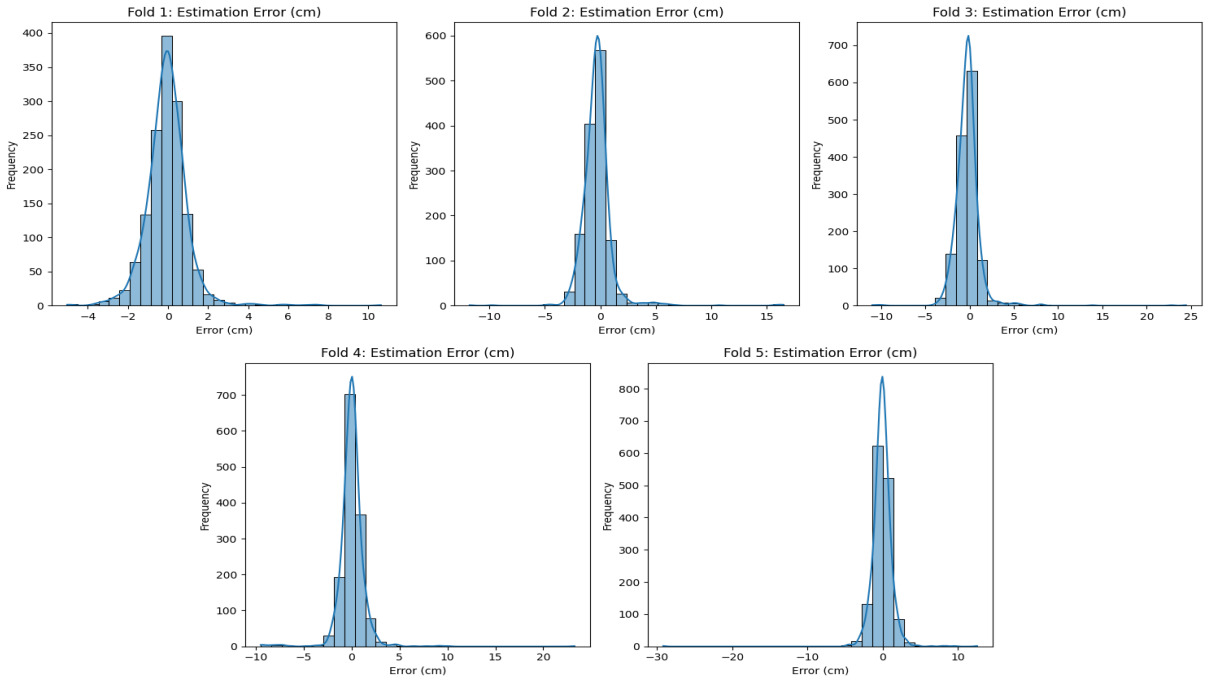


Figure 6.12: Histograms of estimation errors with ESN for X values across five folds.

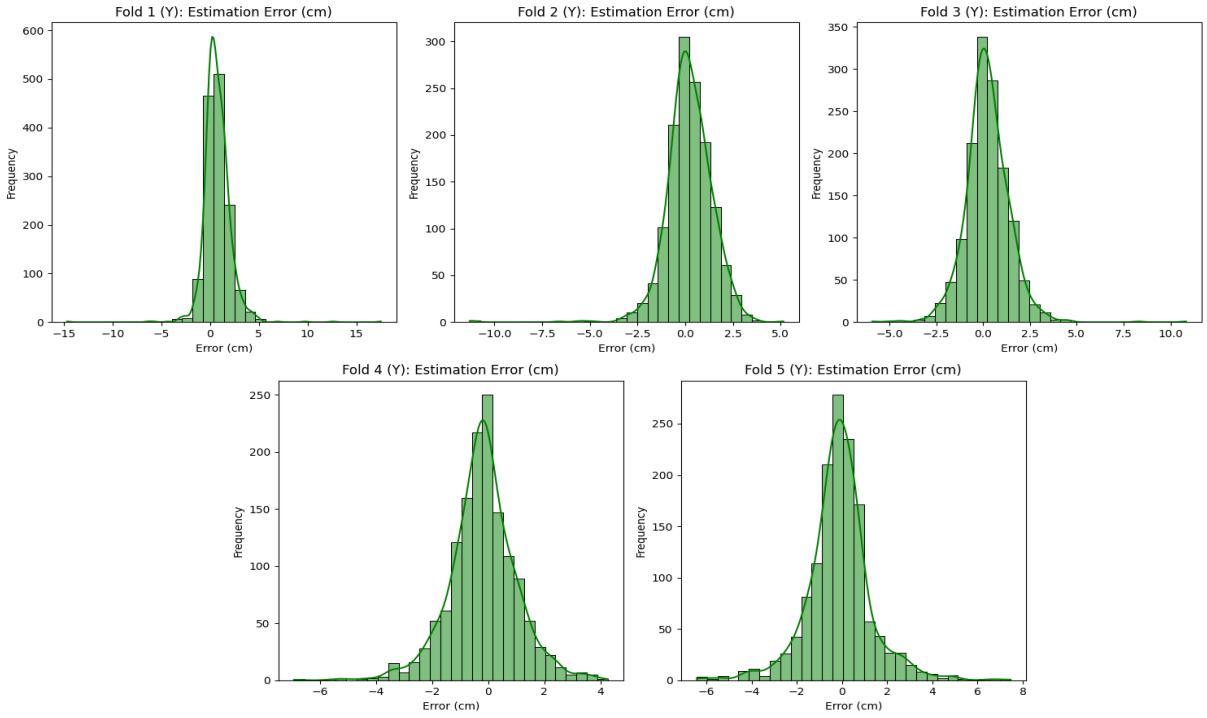


Figure 6.13: Histograms of estimation errors with ESN for Y values across five folds.

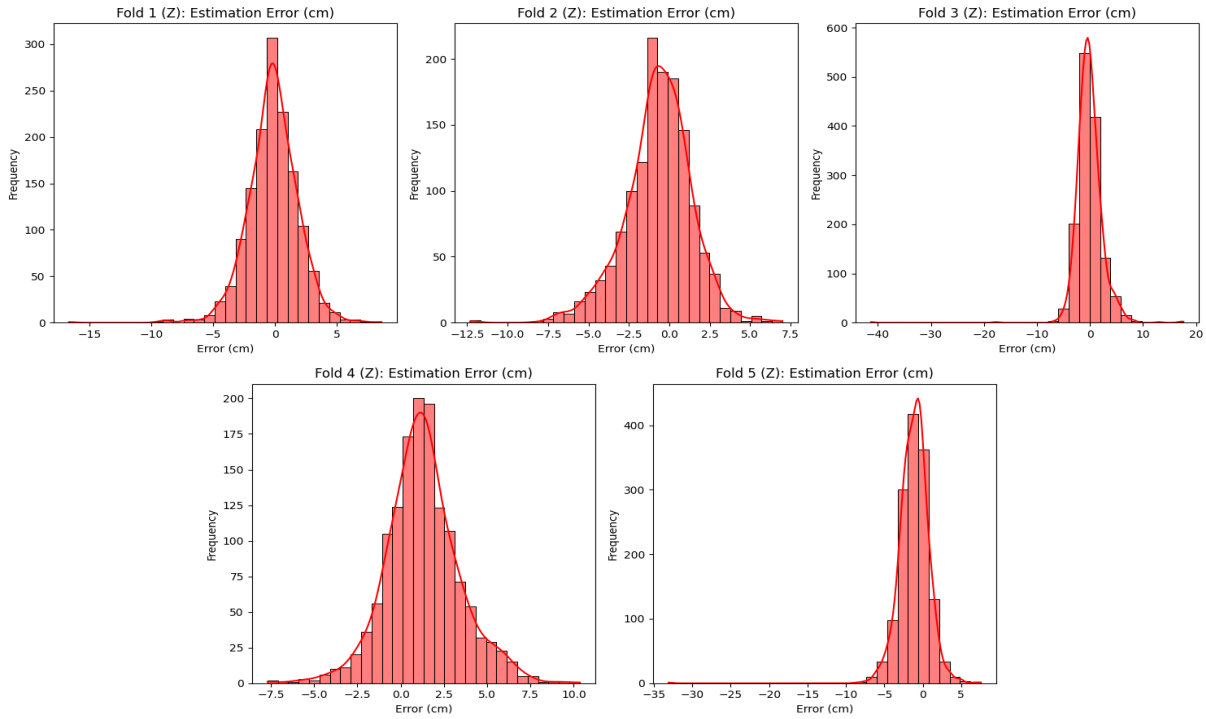


Figure 6.14: Histograms of estimation errors with ESN for Z values across five folds.

The MAE for each network is presented in table 6.5

Table 6.5: ESN Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	0.7231	1.0643	1.5111
Fold 2	0.8729	0.8977	1.6283
Fold 3	0.8955	0.8706	1.6953
Fold 4	0.7974	0.8918	1.9315
Fold 5	0.8521	0.9938	1.6847
Mean ± SD	**0.8282 ± 0.0691**	**0.9436 ± 0.0825**	**1.6902 ± 0.1535**

With respet to the distribution of errors per range, the table 6.6 show the results for the ESN evaluated in the same five fold condition as the CNN.

Table 6.6: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates

Axis	Error Range (cm)	Fold1(%)	Fold2(%)	Fold3(%)	Fold4(%)	Fold5(%)	Mean(%)
X	$(-1, 1)$	76.65	70.97	68.60	76.38	71.18	72.76
	$(-2, 2)$	95.01	92.20	91.85	94.13	92.58	93.15
	$(-3, 3)$	98.38	97.37	97.38	97.67	97.86	97.73
	$(-4, 4)$	99.09	98.32	98.58	98.16	99.00	98.63
Y	$(-1, 1)$	57.81	65.57	66.76	65.77	66.33	64.45
	$(-2, 2)$	87.13	91.17	91.85	90.24	86.52	89.38
	$(-3, 3)$	95.50	98.32	98.09	97.10	94.65	96.73
	$(-4, 4)$	98.10	99.49	99.36	99.43	97.50	98.78
Z	$(-1, 1)$	44.02	42.52	39.76	33.88	38.52	39.74
	$(-2, 2)$	71.87	70.17	69.17	62.52	65.98	67.94
	$(-3, 3)$	88.05	85.70	85.83	79.21	85.73	84.90
	$(-4, 4)$	95.01	92.34	92.63	89.25	93.51	92.55

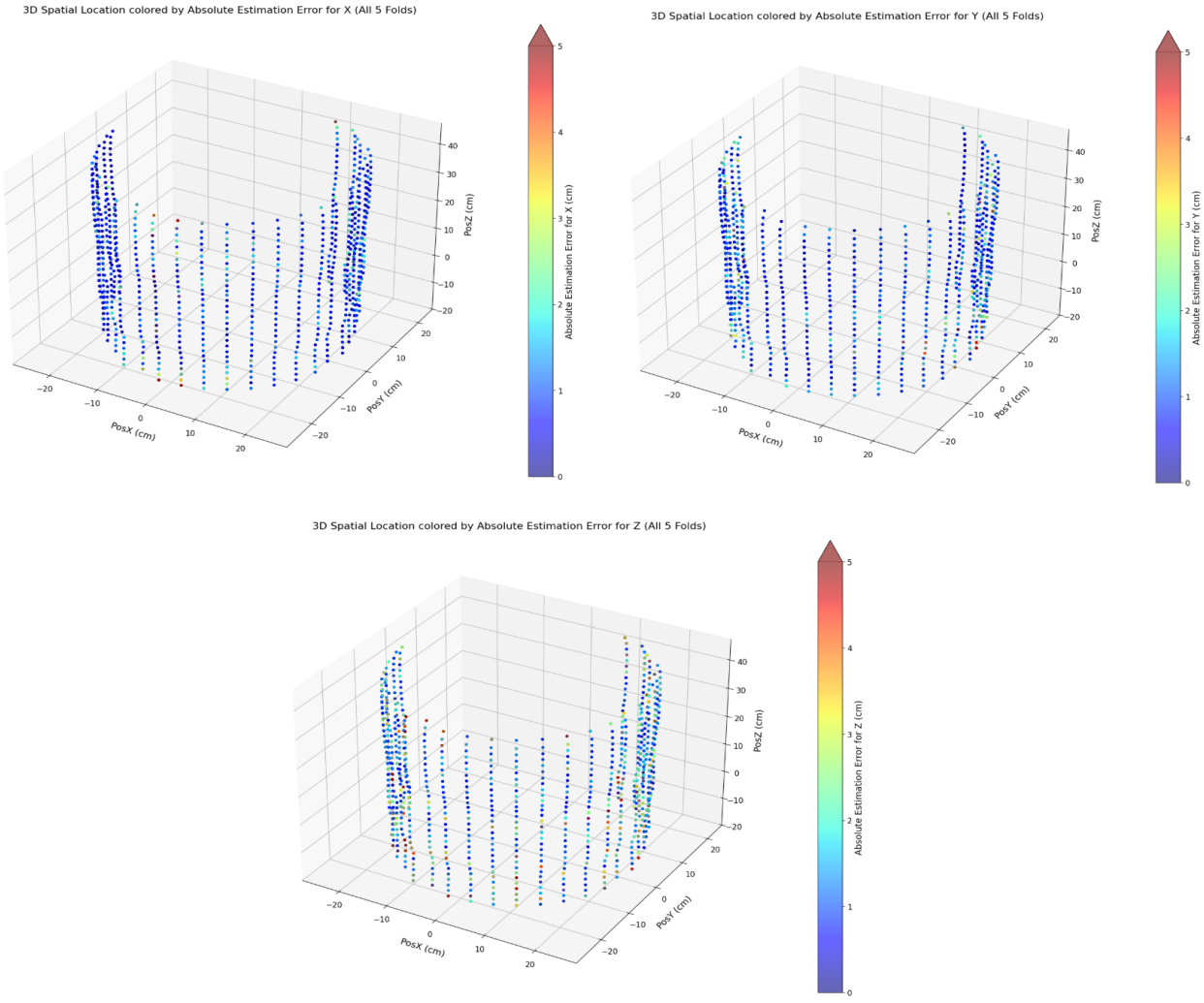


Figure 6.15: 3D scatered plot showing the absolute error value for all the points estimated with the ESN in the five folds as test set.

Table 6.7: Pearson Correlation Coefficients between Ground Truths and Predictions

Folds	X	Y	Z
Fold 1	0.9980	0.9954	0.9929
Fold 2	0.9963	0.9966	0.9929
Fold 3	0.9958	0.9968	0.9898
Fold 4	0.9962	0.9967	0.9921
Fold 5	0.9965	0.9953	0.9936
Mean ± SD	**0.9966 ± 0.0008**	**0.9962 ± 0.0007**	**0.9923 ± 0.0015**

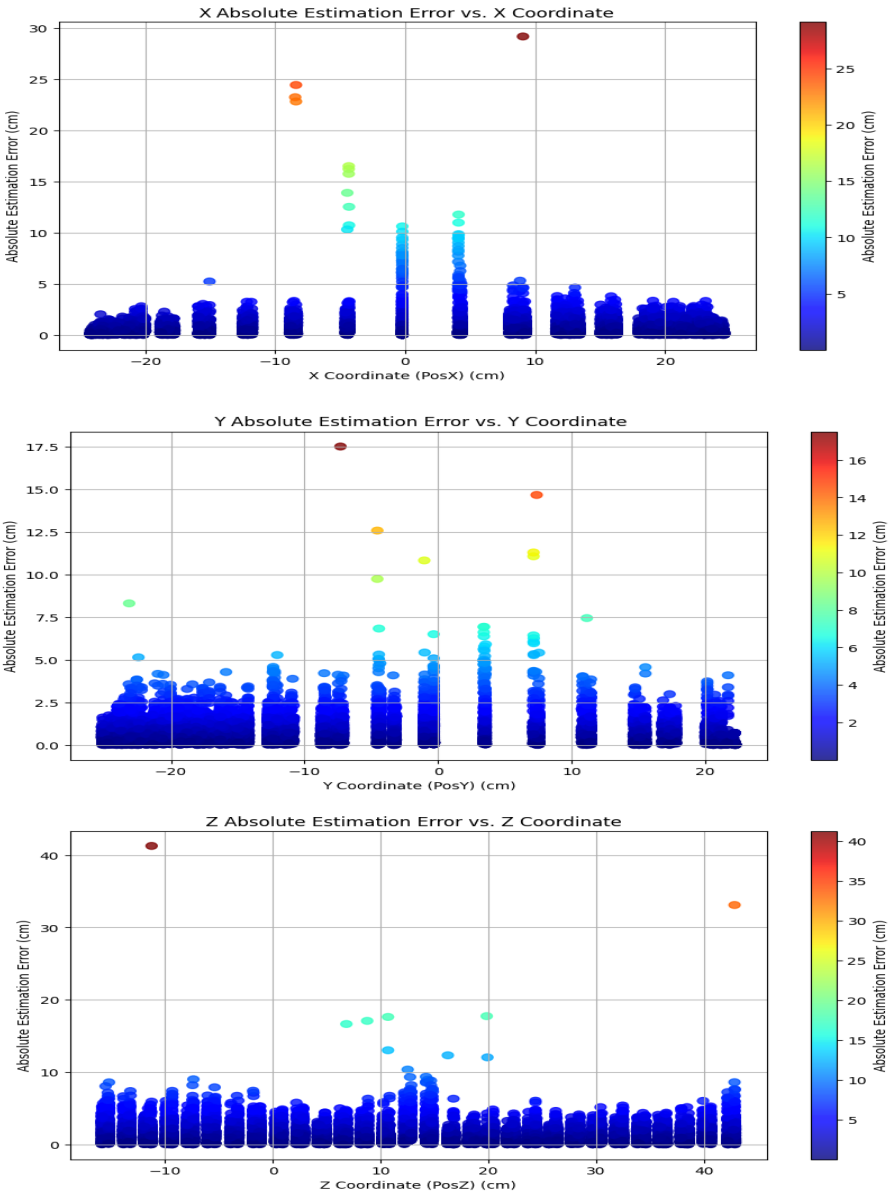


Figure 6.16: Scatered plot showing the absolute estimation error vs its respective coordinate axis for the ESN.

6.5. Beamforming

The following section presents the results for the CNN and ESN models when the input signals are pre-processed with beamforming. The network architectures remain the same as previously described; the only difference is that beamforming is applied to

the input signals to emphasize specific reflections according to the direction from which the acoustic reflection originates.

6.5.1. CNN with Beamforming

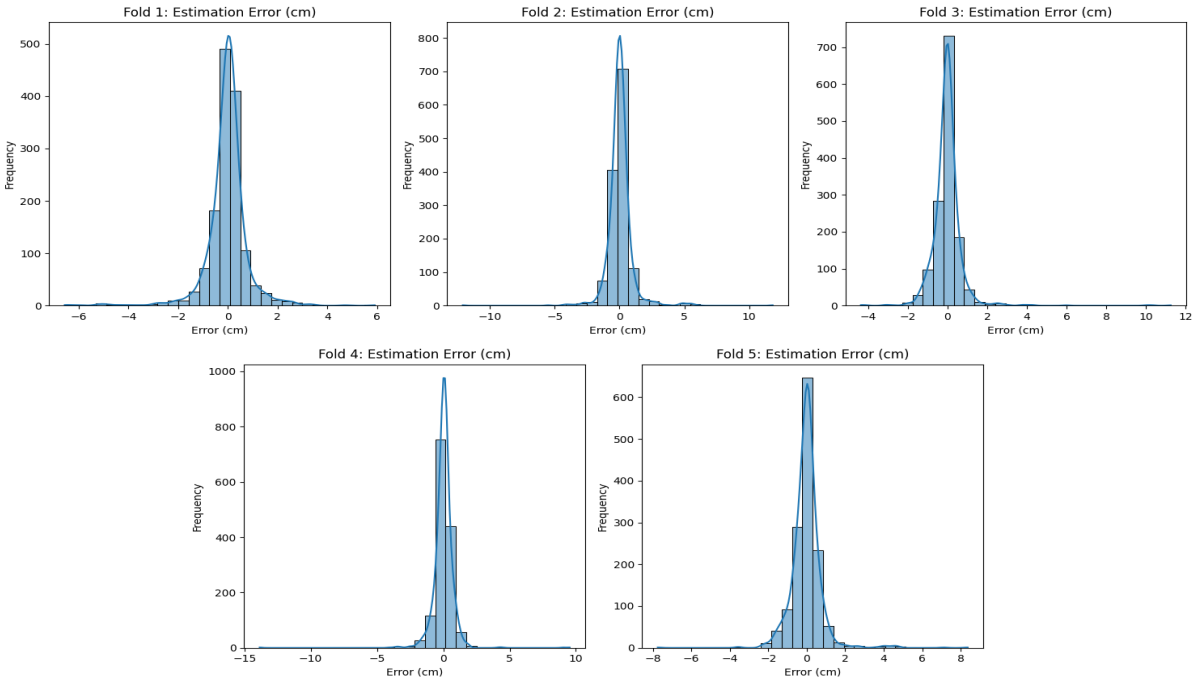


Figure 6.17: Histograms of estimation errors with Beamforming and CNN for X values across five folds.

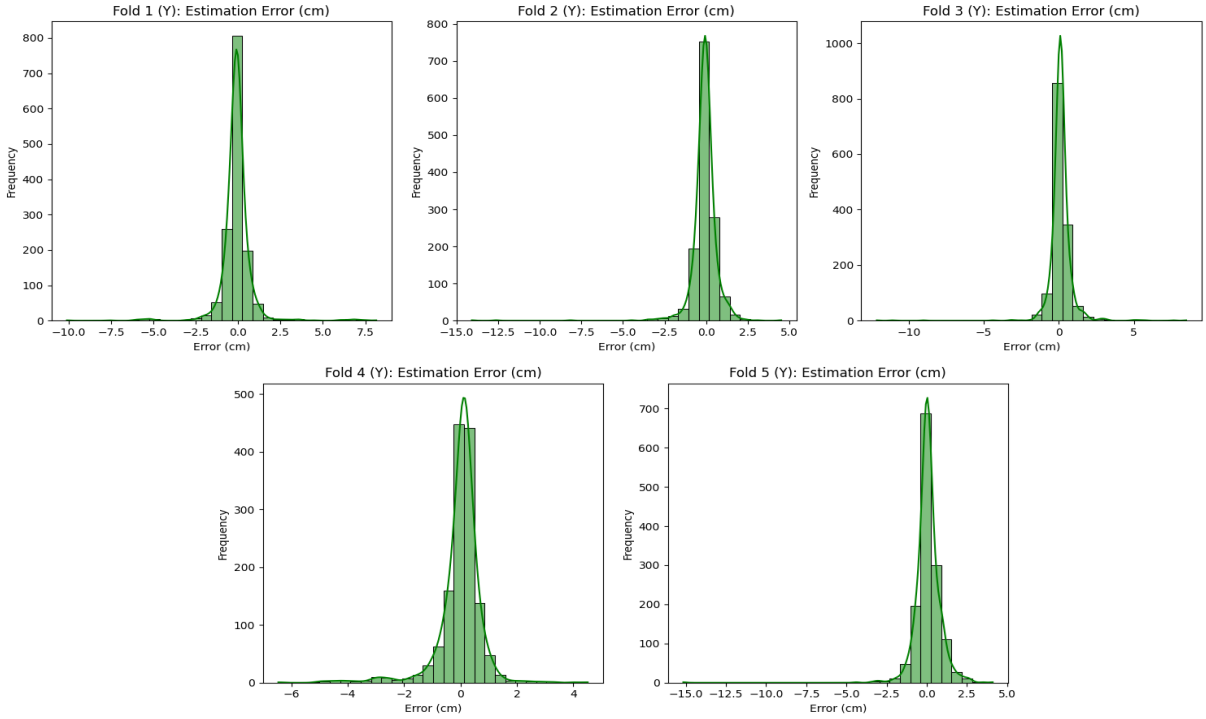


Figure 6.18: Histograms of estimation errors with Beamforming and CNN for Y values across five folds.

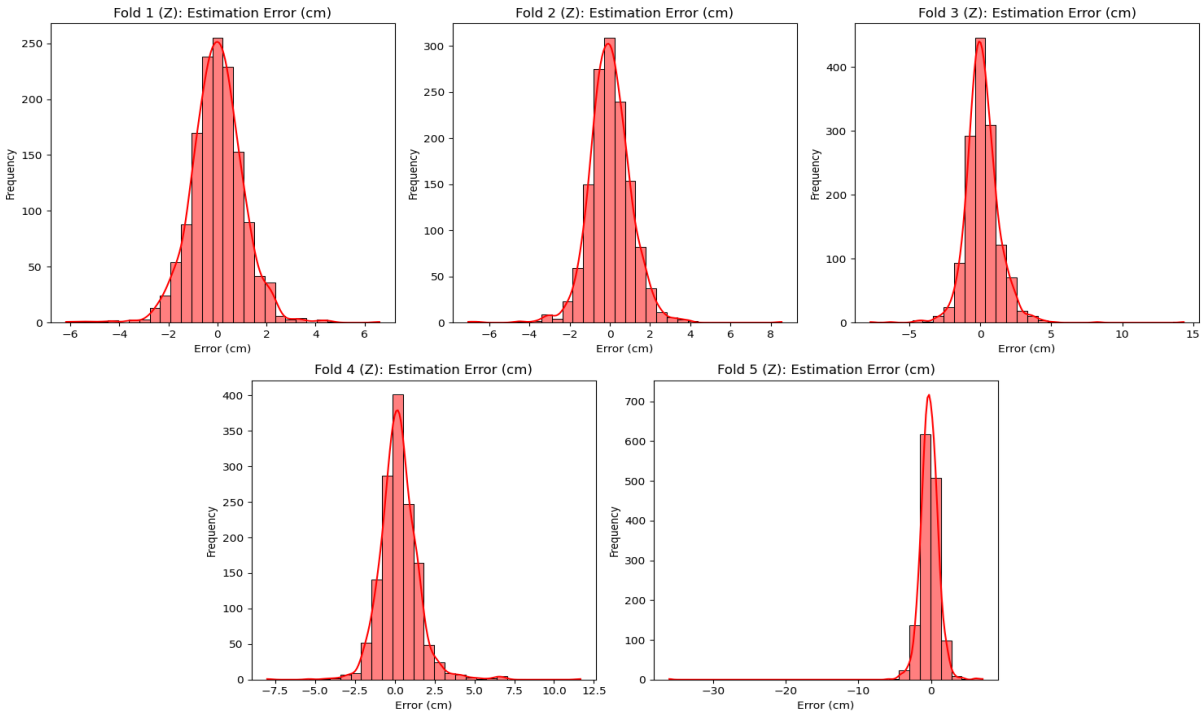


Figure 6.19: Histograms of estimation errors with Beamforming and CNN for Z values across five folds.

The mean absolute error (MAE) for the coordinates across the five folds is presented in table 6.8.

Table 6.8: Mean Absolute Error (MAE) Across Folds for X, Y, and Z Coordinates

Folds	**MAE X (cm)**	**MAE Y (cm)**	**MAE Z (cm)**
Fold 1	0.4941	0.4903	0.8064
Fold 2	0.5467	0.4418	0.7854
Fold 3	0.4246	0.4064	0.8489
Fold 4	0.4426	0.4684	0.8927
Fold 5	0.4865	0.5012	0.9552
Mean ± SD	**0.4789 ± 0.0478**	**0.4616 ± 0.0383**	**0.8577 ± 0.0683**

Other important metrics for the evaluation of the regression model are the percentage of samples with an estimation that falls within a specific error range. The percentage of samples per error range in X, Y and Z is shown in table 6.9.

The estimation error for each point in the dataset is presented in figure 6.20. The

Table 6.9: Percentage of samples with estimation errors within specific ranges for X, Y, and Z Coordinates

Axis	Error Range (cm)	Fold1(%)	Fold2(%)	Fold3(%)	Fold4(%)	Fold5(%)	Mean(%)
X	$(-1, 1)$	87.97	87.02	91.00	90.74	88.16	88.98
	$(-2, 2)$	96.20	96.21	98.23	98.80	97.79	97.44
	$(-3, 3)$	98.66	98.25	99.08	99.29	98.86	98.83
	$(-4, 4)$	99.16	98.69	99.43	99.58	99.22	99.21
Y	$(-1, 1)$	90.08	90.66	92.77	90.17	86.95	90.13
	$(-2, 2)$	96.98	98.18	98.37	96.46	97.50	97.50
	$(-3, 3)$	98.17	99.27	98.94	98.23	99.50	98.82
	$(-4, 4)$	98.45	99.64	99.29	99.08	99.79	99.25
Z	$(-1, 1)$	70.75	72.36	70.45	66.62	63.38	68.71
	$(-2, 2)$	93.53	94.60	91.78	92.72	90.58	92.64
	$(-3, 3)$	98.59	98.47	97.52	97.31	97.43	97.86
	$(-4, 4)$	99.23	99.64	99.08	98.59	98.93	99.09

graph consider the five fold subsets, each at the time it was the test set. The error is calculated as the absolute difference between the predicted and the true value of the coordinates for the respective dimension. The error is represented in centimeters. Figure 6.21 preasent the coordinate error with respect it axis of interest, showing the error estimation for X dimension as a funtion of X location, Y error as fuction of Y and Z error as function of Z.

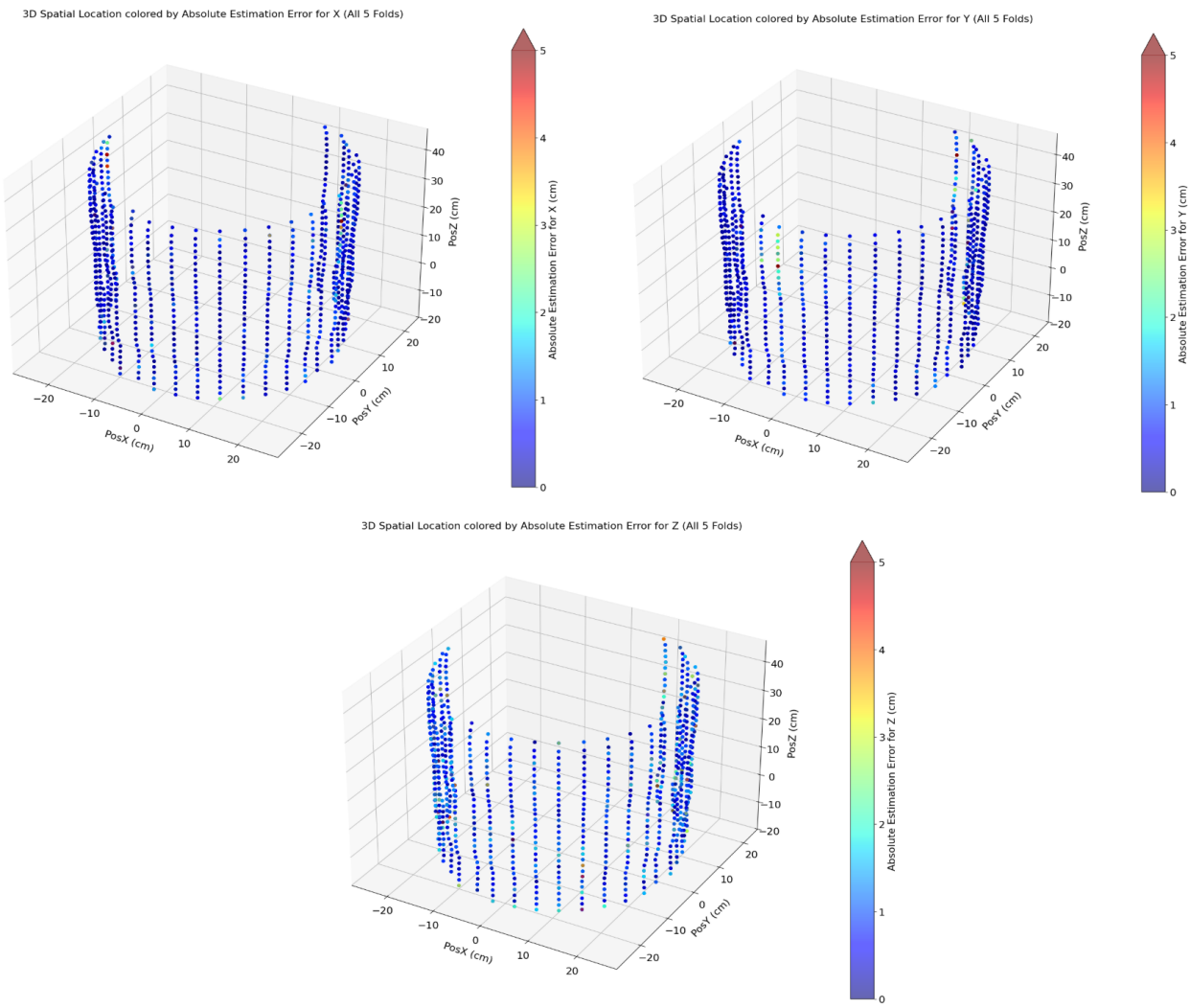


Figure 6.20: 3D scattered plot showing the absolute error value for all the points estimated in the five folds as test set.

To undestand how well the data is correlated for the regresion model, the Pearson correlation coefficient is showns in table 6.10 this coeficient measures the strength and direction of the linear relationship between two continuous variables. This coeficient provide an indication of how well data points cluster around a straight line. In Model Evaluation of Predictions vs. Ground Truth, An Pearson Coeficient > 0.99 indicates that model predictions scale almost perfectly aligned with the actual targets.

Table 6.10: Pearson Correlation Coefficients between Ground Truths and Predictions

Folds	X	Y	Z
Fold 1	0.9966	0.9940	0.9980
Fold 2	0.9952	0.9952	0.9980
Fold 3	0.9966	0.9949	0.9979
Fold 4	0.9967	0.9953	0.9973
Fold 5	0.9967	0.9956	0.9962
Mean ± SD	**0.9964 ± 0.0007**	**0.9950 ± 0.0006**	**0.9975 ± 0.0008**

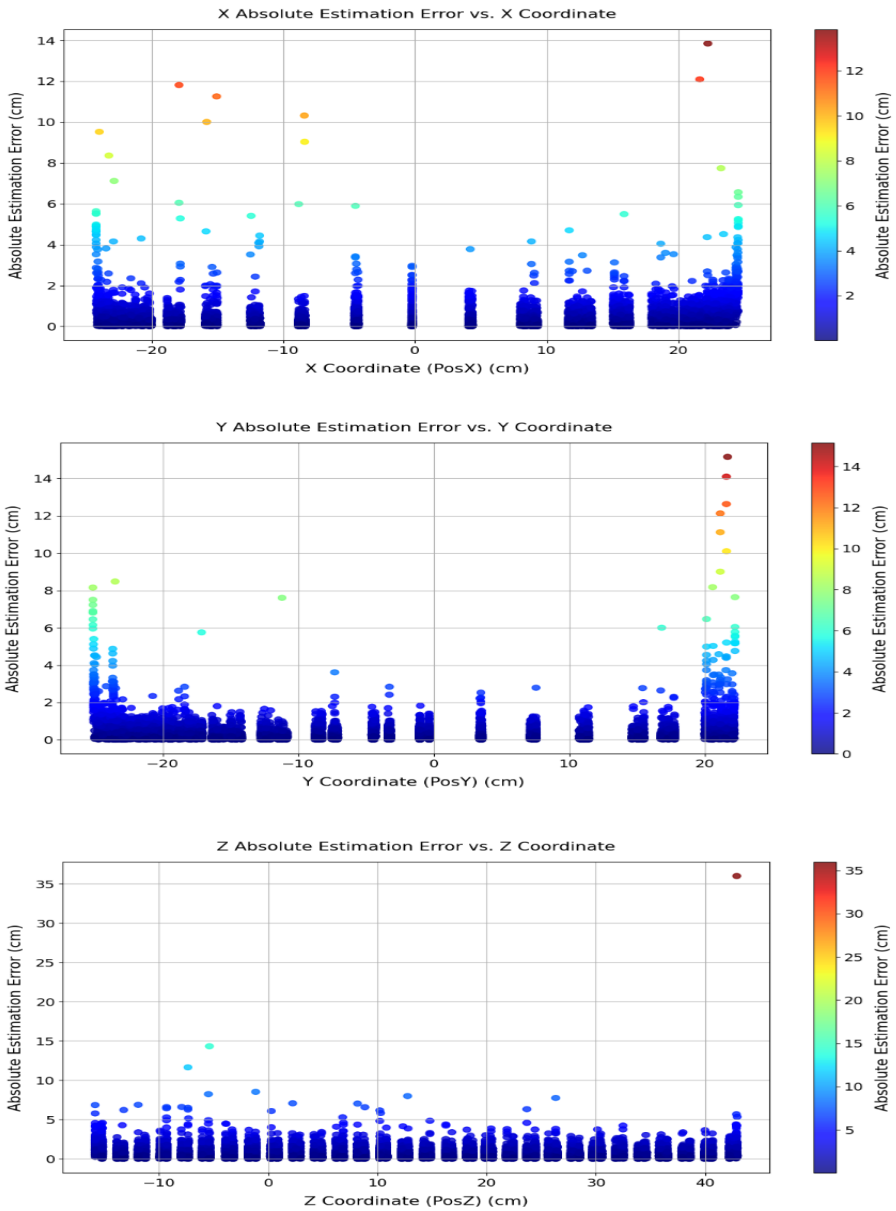


Figure 6.21: Scattered plot showing the absolute estimation error vs the respective coordinate axis.

6.5.2. ESN with Beamforming

Additional metrics obtained during the development of the research are presented in this section.

6.6. Correctness (if needed)

If the correctness of the proposal was evaluated, it should be presented in this section.

6.7. Output (If needed)

If the proposal's output differs from what was analyzed with the statistical tool, it should be presented in this section.

Chapter 7

Discussion

"Another quote."

Another Author

7.1. CNN model and the regression problem

7.1.1. CNN model to identify cube regions and orientations

Following the problem-complexity tiers described by Vapnik in [40], the model was first evaluated as a classifier with the task of determine the cube's location within some regions and the cube orientation. For the location-classification task across the acoustic sessions, the CNN achieved 98.59% accuracy on the test set. This result indicates that the audio signals, once transformed into spectrograms, contain sufficient information for spatial localization and that the network can extract the features needed to identify the cube's region within the measurement box. The confusion matrix in Figure 6.1 shows a clear diagonal pattern, indicating strong classification performance. Twenty of the 26 classes achieved 100% recall. Most errors occurred between neighboring classes; class 1 had the highest number of misclassifications, with 13 samples classified as class 0, resulting in 75% accuracy for that particular class.

In the orientation-classification task, the network was required to assign the cube's orientation to one of nine classes and achieved 29.06% accuracy on the test set. Unlike the location-classification results, the confusion matrix in Figure 6.2 does not show a clear diagonal pattern. Instead, it reveals three clusters of misclassifications near the diagonal. In the first cluster, samples from classes 0, 1, and 2 are often classified as class 1. In the middle cluster, samples from classes 2, 3, 4, and 5 are often classified as class 4. In the rightmost cluster, samples from classes 5, 6, 7, and 8 are often classified as class 6. Overall, predictions are concentrated in classes 1 (10 degrees), 4 (40 degrees), and 6 (60 degrees). This pattern suggests that the network tends to favor a few broad orientation categories spaced by arround 20 to 30-degree rather than reliably distinguishing all nine classes, which represent 10-degree increments.

The further exploration to analize the efect of the proximity of the cube to the microphones and the relative location of its front face with respect to the sides of the

measure box, the acuracy by regions shows in figure 6.1 tells that the performance is not improved by the proximity or the reducton of variability with respect of the cube location with respect to it location inside the measurement box. The fact that the accuracy yield even lower can be explain by the reduction of samples avaliable for training in those reduced regions. One aspect to highlight for the experiment of clasification into the subregios is that even though the acurracy were low, the confusion matrix of figure 6.3 shows better distribution of samples along the matrix diagonal.

Comparing the two tasks, the results show a marked difference in performance: spatial-location classification achieved over 98% accuracy, whereas orientation classification achieved only 29.06%. This difference may arise because rotating the cube changes the orientation of its surfaces without substantially changing their positions relative to the microphones and the wall in the box. Consequently, orientation may be encoded in subtle inter-channel phase shifts or in the attenuation of certain echoes, rather than in pronounced Time-Difference-Of-Arrival (TDOA) cues. By contrast, translating the cube changes the source–microphone geometry and may produce more distinct TDOA cues and changes in the reverberation pattern. The CNN may therefore have difficulty isolating orientation-dependent features in the spectrograms, which could explain why its predictions are concentrated in a few orientation classes.

7.1.2. CNN model to estimate the cube location

In the case of estimating the cube's location for the individual coordinates X, Y and Z using the CNN model, the figures 6.6, 6.7, and 6.8 present the histograms of the prediction errors for the X, Y, and Z coordinates obtained from the five folds. The histograms indicate that the model is generally able to localize the cube with reasonable accuracy, with most of the errors falling within a narrow range of values arround ± 2 cm, narrow relative to the dimension of the measurement box and the cube size. Most errors

are concentrated around zero, suggesting that the network's predictions are close to the true positions. However, there are occasional larger errors, which may be attributed to challenging acoustic conditions like noise during recording. The X and Y histograms show comparable results, for Z coordinate estimation the fold 1 and fold 4 present wider distributions with more errors above the range of ±2 cm that indicated the Z estimation is more challenging for the network to estimate the coordinates. Overall, the CNN demonstrates a strong capability for spatial localization, consistent with the high classification accuracy observed in the previous section, indicating that the CNN is able to accurately estimate the cube's position in all three dimensions.

The MAE is the main figure of merit to evaluate the estimation capability of the models [29]. The table 6.2 summarizes the mean and standard deviation of the prediction errors for the X, Y, and Z coordinates across the five folds. The results indicate that the CNN achieves the lowest errors for Y coordinates, followed by X. For these two coordinates the network performs similarly well, with MAEs arround 0.5 cm. In the case of Z the mean MAE almost double the values for X and Y with a MAE of 0.9 cm and bigger standard deviation. The results in the table are consistent with the observations from the histograms. This further supports the conclusion that the network is more accurate in estimating the horizontal positions of the cube than its vertical position.

To understand the distribution of errors by tiers, the table 6.4 present the percentage of samples in the dataset with estimation error within a especific error range. The results are congruent with the previous paragraph: Y is the most accurately estimated coordinate: 92.90% of samples fall within ±1 cm of error, reaching 98.32% within ±2 cm on average across the 5 folds. X exhibits high precision with 87.69% of samples fall within ±1 cm, rising rapidly to value comparable to Y of 97.83% within ±2 cm. Z presents the highest fine-scale error and inter-fold variance: Only 64.44% of samples achieve estimation errors within ±1 cm, and that the range of ±3 cm. is the one to

have a comparable value of 98.32%, that presenting this network 8.92% of errors values above ±2 cm. For the three CNN estimators, errors exceeding ±4 cm represent less than 0.6% of all samples (X: 99.42%, Y: 99.69%, Z: 99.45%).

Other figure of merit to evaluate regresion is the Pearson Correlation Coefficients [29] shown in table 6.10 for all the folds in the three coordinates with the exeption of fold 3 on Z coordinate (a value of 0.9898), the coeficient is greater than 0.99 for the rest of the cases, indicating a strong linear relationship between the predicted and true coordinates. This further confirms the network's capability to accurately estimate the cube's position in the measurement box.

The error distributions across the three dimensions are illustrated in Figure 6.9, which shows the errors in the X, Y, and Z coordinates throughout the measurement box. The visualization indicates that errors are distributed relatively uniformly across the measurement space. Most errors are close to zero, while larger errors occur in a few regions; however, no clear clustering pattern is apparent. The figure 6.10 present the error for each coordinate only as function of it own cordinate value. In the case of X absolute estimate errors as a function of the X coordinate can be observe that the error remains low (near 2cm) in the central area and increase near the edges, with the boundary concentrating the estimation outliers. For Y a similar behaviour is observed, the center of the axis present lower values than the edge, but the error values in the center are even lower than the obtained in the estimation of X, this suggest that the task of estimating the coordinate on the Y axis is easier for the network compared to estimating magnitudes in the X domain. This aspect is particularly interesting due the fact that for the X estimation, the network is trained with a binaural input channels what allow the network to exploit the physical separation between two sound receivers, capable to learn interaural time difference from the channels, but the Y estimation perform bettern than the X. The fact that Y estimation have a strong relationship to the distance the cube is located in the path from the acoustic signal from the speaker

source to the receiver microphones and this add a distintive interference effect when the cube is near the speakers or near the microphones that could explaing why the estimation along this axis present lower errors than X. For the estimation of Z, the errors are higher than X and Y. One possible explanation is that the Z-coordinate values are not uniformly distributed around zero: the dataset contains more positive values than negative ones. Consequently, the network is trained on more samples in the range from zero to 40 than in the negative range. This imbalance may contribute to the larger estimation errors observed for negative Z values. In addition, some negative Z values lie in a region occupied by the robotic arm's base, which may further complicate the estimation in that part of the measurement space. two general observation

7.2. ESN model and the regression problem

As with the CNN, the ESN was first evaluated as a classifier to assess its ability to extract from the acoustic data the features needed to locate the cube within the measurement box. The ESN achieved 98.42% accuracy, just 0.17 percentage points below the CNN's 98.59%. The confusion matrix in Figure 6.11 shows a well-defined diagonal, indicating close agreement between the true and predicted labels. The few misclassifications are between neighboring classes. Unlike the CNN, the ESN also classifies sessions 0 and 1 more accurately; these sessions correspond to locations at the physical edge of the measurement area.

As with the CNN, the error distributions for the ESN estimators are shown in Figures 6.12, 6.13, and 6.14 for the X, Y, and Z coordinates, respectively. As in the CNN results, the histograms have approximately bell-shaped distributions centered around zero. However, the ESN error distributions are wider, indicating greater variability than that observed with the CNN.

The MAE values reported for the ESN in Table 6.5 are higher overall than those

reported for the CNN in Table 6.2. However, the relative performance of the X and Y estimators is reversed: the ESN achieves a lower MAE for X than for Y, while Z has the highest MAE of the three coordinates. The ESN's sensitivity to hyperparameter selection may partly explain these differences. Because the hyperparameters were initially tuned for regression on X, they may not have been optimal for Y, despite both coordinates using the same ground-truth transformation. The Z-coordinate data also have a different spatial distribution, and their ground-truth values do not require the arcsine normalization described in Equation 5.2. These differences may further affect how well the selected hyperparameters generalize across ESN for diferent coordinates.

The error percentages reported across different tolerance ranges in Table 6.6 are consistent with the wider ESN error distributions shown in the histograms. A tolerance of ± 2 cm captures a smaller proportion of the estimates; a tolerance of ± 3 cm is needed to capture approximately 97% of them.

The Pearson correlation coefficients on table 6.7 also exceed 0.99 in most cases. The only exception is the Z estimator in fold 3, suggesting that the regression model may have difficulty capturing the relationship between the features and target values for some samples in this fold.

Figure 6.15 shows the spatial distribution of the estimation errors across all three dimensions. The scatter plot indicates that the error magnitudes are distributed relatively uniformly throughout the measurement region. However, the coordinate-wise plots in Figure 6.16 show that the largest errors occur in a specific interior region rather than near the limits of the measurement space. This behavior may be explained by the multi-head architecture used for coordinate estimation. Two regression heads estimate the cube's position in the negative and positive coordinate ranges, respectively, while a third head selects which estimate to use. This selection may be less reliable near the center, where the two ranges meet. A fourth head

identifies outliers by comparing the estimates; however, if the discrepancy between them is below the specified threshold, an incorrect head selection may not be flagged. This could explain why the central region remains prone to errors. Training both regression heads on an overlapping region around the boundary could mitigate this issue: each head would then learn from samples on both sides of the boundary, reducing the error when the selector chooses the less appropriate head.

7.3. Beamforming

7.3.1. CNN with Beamforming

7.3.2. ESN with Beamforming

7.4. Model Comparison

This indicates that while the ESN can still provide reasonably accurate estimates, it is more prone to larger errors compared to the CNN. Consequently, for applications where minimizing the largest errors is critical, the CNN may be the preferred choice despite its higher computational cost.

Despite the higher MAE values, the ESN offers a significant reduction in model complexity compared to the CNN. The ESN requires only about 3% of the trainable parameters used by the ResNet18 CNN, making it a more efficient option for the regression task with acoustic features as input. This efficiency could be particularly advantageous in scenarios with limited computational resources or when real-time processing is required. Overall, while the CNN achieves lower absolute errors, the ESN provides a compelling trade-off between accuracy and computational efficiency.

Chapter 8

Conclusions and Future Work

"Another quote."

Another author

This chapter presents the final conclusions obtained through the research effort and the future work that can be derived.

8.1. Conclusions

- Conclusion 1.
- Conclusion 2.
- Conclusion 3.

8.2. Future Work

- Future work 1.
- Future work 2.
- Future work 3.

Chapter 9

Appendixes

9.1. Source Code

Pointer or description of the source code can be given in this section..

9.2. Gathered Data from the Experiments

If required, raw data could be presented in this section.

References

[1] Ossama Abdel-Hamid, Abdel-rahman Mohamed, Hui Jiang, Li Deng, Gerald Penn, and Dong Yu. Convolutional neural networks for speech recognition. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 22(10):1533–1545, Oct 2014. ISSN 2329-9290. doi: 10.1109/TASLP.2014.2339736.

[2] Sharath Adavanne, Archontis Politis, Joonas Nikunen, and Tuomas Virtanen. Sound event localization and detection of overlapping sources using convolutional recurrent neural networks. *IEEE Journal of Selected Topics in Signal Processing*, 13(1):34–48, 2019. doi: 10.1109/JSTSP.2018.2885636.

[3] Rosepreet Kaur Bhogal and Ajmer Singh. A comprehensive review of resnet-18: Architecture and applications. In Jagdish Chand Bansal, Snehanshu Saha, Carlos A. Coello Coello, and Hemant Rathore, editors, *Advances in Data-Driven Computing and Intelligent Systems*, pages 595–607, Singapore, 2025. Springer Nature Singapore. ISBN 978-981-96-4536-7.

[4] Jacob Bitterman, Daniel Levi, Hilel Hagai Diamandi, Sharon Gannot, and Tal Rosenwein. Revrir: Joint reverberant speech and room impulse response embedding using contrastive learning with application to room shape classification, 2024. URL https://arxiv.org/abs/2406.03120.

[5] Amandine Brunetto, Sascha Hornauer, Stella X. Yu, and Fabien Moutarde. The audio-visual batvision dataset for research on sight and sound. In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, page 1–8. IEEE, October 2023. doi: 10.1109/iros55552.2023.10341715. URL http://dx.doi.org/10.1109/IROS55552.2023.10341715.

[6] Laurens Byttebier. *Small Footprint Neural Network Models for Acoustic Scene Classification.* Mtech thesis, Ghent University, 2021. Available at https://libstore.ugent.be/fulltxt/RUG01/003/014/483/RUG01-003014483_2021_0001_AC.pdf.

[7] Ludovick Lepauloux Daniele Battaglino and Nicholas Evans. Acoustic scene classification using convolutional neural networks, 2016. URL https://dcase.community/documents/challenge2016/technical_reports/DCASE2016_Battaglino_1032.pdf.

[8] Leandro de Santana. Fundamentals of acoustic beamforming, 2017. URL https://www.sto.nato.int/publications/STO%20Educational%20Notes/STO-EN-AVT-287/EN-AVT-287-04.pdf.

[9] Hugo Demontis, Francois Ollivier, and Jacques Marchal. 3d identification of acoustic sources in rooms using a large-scale microphone array. In *2018 16th International Workshop on Acoustic Signal Enhancement (IWAENC)*, pages 506–509, 2018. doi: 10.1109/IWAENC.2018.8521264.

[10] Maria Deprez and Emma C. Robinson. Chapter 11 - convolutional neural networks. In Maria Deprez and Emma C. Robinson, editors, *Machine Learning for Biomedical Applications*, pages 233–270. Academic Press, 2024. ISBN 978-0-12-822904-0. doi: https://doi.org/10.1016/B978-0-12-822904-0.00016-9. URL https://www.sciencedirect.com/science/article/pii/B9780128229040000169.

[11] Ivan Dokmanić, Reza Parhizkar, Andreas Walther, Yue M. Lu, and Martin Vetterli. Acoustic echoes reveal room shape. *Proceedings of the National Academy of Sciences*, 110(30):12186–12191, 2013. doi: 10.1073/pnas.1221464110. URL https://www.pnas.org/doi/abs/10.1073/pnas.1221464110.

[12] Ivan Dokmanić, Reza Parhizkar, Andreas Walther, Yue M. Lu, and Martin Vetterli. Acoustic echoes reveal room shape. *Proceedings of the National Academy of Sciences*, 110(30):12186–12191, 2013. doi: 10.1073/pnas.1221464110. URL https://www.pnas.org/doi/abs/10.1073/pnas.1221464110.

[13] Johann Faouzi. Time series classification: A review of algorithms and implementations. In Jorge Rocha, Cláudia M. Viana, and Sandra Oliveira, editors, *Time Series Analysis*, chapter 2. IntechOpen, Rijeka, 2024. doi: 10.5772/intechopen.1004810. URL https://doi.org/10.5772/intechopen.1004810.

[14] Òscar Garibo-i Orts, Nicolas Firbas, Laura Sebastiá, and J. Alberto Conejero. Gramian angular fields for leveraging pretrained computer vision models with anomalous diffusion trajectories. *Phys. Rev. E*, 107:034138, Mar 2023. doi: 10.1103/PhysRevE.107.034138. URL https://link.aps.org/doi/10.1103/PhysRevE.107.034138.

[15] Letizia Gionfrida, Chen Wang, Lu Gan, Margarita Chli, and Luca Carlone. Computer and robot vision: Past, present, and future [tc spotlight]. *IEEE Robotics & Automation Magazine*, 31(3):211–215, 2024. doi: 10.1109/MRA.2024.3428780.

[16] Yoonchang Han, Jeongsoon Park, and Kyogu Lee. Convolutional neural networks with binaural representations and background subtraction for acoustic scene

classification. In *Workshop on Detection and Classification of Acoustic Scenes and Events*, 2017. URL https://api.semanticscholar.org/CorpusID:52830611.

[17] Gunnar Heilmann, Andy Meyer, and Dirk Döbler. Beamforming in the time-domain using 3d-microphone arrays. In *XIXth Biennial Conference of the New Zealand Acoustical Society, Auckland*, 2008.

[18] Julien I.E. Hoffman. Chapter 26 - analysis of variance. ii. more complex forms. In *Basic Biostatistics for Medical and Biomedical Practitioners (Second Edition)*, pages 419–441. Academic Press, second edition edition, 2019. ISBN 978-0-12-817084-7. doi: https://doi.org/10.1016/B978-0-12-817084-7.00026-7. URL https://www.sciencedirect.com/science/article/pii/B9780128170847000267.

[19] Shengli Jiang, Shiyi Qin, Joshua L. Pulsipher, and Victor M. Zavala. Chapter 3 - convolutional neural networks: Basic concepts and applications in manufacturing. In Masoud Soroush and Richard D Braatz, editors, *Artificial Intelligence in Manufacturing*, pages 63–102. Academic Press, 2024. ISBN 978-0-323-99134-6. doi: https://doi.org/10.1016/B978-0-323-99134-6.00007-4. URL https://www.sciencedirect.com/science/article/pii/B9780323991346000074.

[20] Shaowei Jin, Huaping Liu, Bowen Wang, and Fuchun Sun. Open-environment robotic acoustic perception for object recognition. *Frontiers in Neurorobotics*, Volume 13 - 2019, 2019. ISSN 1662-5218. doi: 10.3389/fnbot.2019.00096. URL https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2019.00096.

[21] Jacqueline Koeppen and Gunnar Heilmann. 3d beam forming measurements using 3d-microphone arrays. Technical report, SAE Technical Paper, 2009.

[22] Teodoro Aguilera Leticia Jaén, Fernando J. Álvarez and Juan J. García. Groom-level indoor positioning based on acoustic impulse response identification. *International Conference on Indoor Positioning and Indoor Navigation*, September 2017.

[23] Mantas Lukoševičius and Herbert Jaeger. Reservoir computing approaches to recurrent neural network training. *Computer Science Review*, 3(3):127–149, 2009. ISSN 1574-0137. doi: https://doi.org/10.1016/j.cosrev.2009.03.005. URL https://www.sciencedirect.com/science/article/pii/S1574013709000173.

[24] Shan Luo, Leqi Zhu, Kaspar Althoefer, and Hongbin Liu. Knock-knock: Acoustic object recognition by using stacked denoising autoencoders. *Neurocomputing*, 267:18–24, 2017. ISSN 0925-2312. doi: https://doi.org/10.1016/j.neucom.2017.03.014. URL https://www.sciencedirect.com/science/article/pii/S092523121730509X.

[25] Tomoya Manabe, Rikuto Fukunaga, Kei Nakatsuma, and Makoto Kumon. Object surface recognition using microphone array by acoustic standing wave. In *2022*

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 9274–9279, 2022. doi: 10.1109/IROS47612.2022.9981386.

[26] Iain McCowan. Microphone arrays : A tutorial. In *Computer Science, Engineering*, 2001. URL https://api.semanticscholar.org/CorpusID:16623291.

[27] mordorintelligence. Usa maintenance repair & operations industry size & share analysis - growth trends & forecasts (2025 - 2030), 2025. URL %https://www.mordorintelligence.com/industry-reports/united-states-maintenance-repair-and-operations-mro-market.

[28] Nils Peters, Howard Lei, and Gerald Friedland. Name that room: room identification using acoustic features in a recording. MM '12, page 841–844, New York, NY, USA, 2012. Association for Computing Machinery. ISBN 9781450310895. doi: 10.1145/2393347.2396326. URL https://doi.org/10.1145/2393347.2396326.

[29] Mauri Rainio, Aleksi Teuho, and Riku Klén. Evaluation metrics and statistical tests for machine learning. *Scientific Reports*, 14:6038, 2024. doi: 10.1038/s41598-024-56706-x. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC10937649/.

[30] Michael Rothstein, Zhuolin Liu, and Islam Bashima. Echosense: Acoustic sensing for enclosed structure monitoring and recognition. In *2023 IEEE MIT Undergraduate Research Technology Conference (URTC)*, pages 1–6, 2023. doi: 10.1109/URTC60662.2023.10534944.

[31] Faizan Samad. Using audio ray tracing to calculate room impulse response, 2009. URL https://courses.washington.edu/phys536/presentations/mar%2009/samad-Applications%20of%20Audio%20Ray%20Tracing%20(1).pdf.

[32] CAE Software, Systems, and 01db. Noise inspector - acoustic cameras technical datasheet, 2016. URL https://www.gkmconsultants.com/assets/documents/instruments/pdf/not1542-june-2016-e-noise-inspector-acoustic-camera-technical-data-sheet-en.pdf.

[33] Jasna Stamać, Sanja Grubeša, and Antonio Petošić. Designing the acoustic camera using matlab with respect to different types of microphone arrays. In *2019 2nd International Colloquium on Smart Grid Metrology (SMAGRIMET)*, pages 1–6, 2019. doi: 10.23919/SMAGRIMET.2019.8720363.

[34] Winter 2007 stanford edu, Music 318. Room impulse response measurement and analysis, 2007. URL https://ccrma.stanford.edu/~adnanm/SCI220/Music318ir.pdf.

[35] Chenxi Sun, Moxian Song, Shenda Hong, and Hongyan Li. A review of designs and applications of echo state networks. *CoRR*, abs/2012.02974, 2020. URL https://arxiv.org/abs/2012.02974.

[36] Iuliana Tabian, Hailing Fu, and Zahra Sharif Khodaei. A convolutional neural network for impact detection and characterization of complex composite structures. *Sensors*, 19(22), 2019. ISSN 1424-8220. doi: 10.3390/s19224933. URL https://www.mdpi.com/1424-8220/19/22/4933.

[37] Ethan Tracy and Navinda Kottege. Catchatter: Acoustic perception for mobile robots. *IEEE Robotics and Automation Letters*, 6(4):7209–7216, 2021. doi: 10.1109/LRA.2021.3094492.

[38] Van-Thuan Tran, Chia-Yang Wu, and Wei-Ho Tsai. Acoustic-based lego recognition using attention-based convolutional neural networks. *Artif. Intell. Rev.*, 57(1), January 2024. ISSN 0269-2821. doi: 10.1007/s10462-023-10625-x. URL https://doi.org/10.1007/s10462-023-10625-x.

[39] Spoorthy. V, Manjunath Mulimani, and Shashidhar G. Koolagudi. Acoustic scene classification using deep learning architectures. In *2021 6th International Conference for Convergence in Technology (I2CT)*, pages 1–6, 2021. doi: 10.1109/I2CT51068.2021.9418177.

[40] Vladimir N. Vapnik. *Statistical Learning Theory*. Adaptive and Learning Systems for Signal Processing, Communications, and Control. John Wiley & Sons, New York, NY, USA, 1998. ISBN 978-0-471-03003-4.

[41] Víctor Manuel Vargas, Rafael Ayllón-Gavilán, Antonio Manuel Durán-Rosal, Pedro Antonio Gutiérrez, César Hervás-Martínez, and David Guijo-Rubio. Gramian angular and markov transition fields applied to time series ordinal classification. In Ignacio Rojas, Gonzalo Joya, and Andreu Catala, editors, *Advances in Computational Intelligence*, pages 505–516, Cham, 2023. Springer Nature Switzerland. ISBN 978-3-031-43078-7.

[42] S. Velasco, F. L. Román, A. González, and J. A. White. A computer-assisted experiment for the measurement of the temperature dependence of the speed of sound in air. *American Journal of Physics*, 72(2):276–279, 02 2004. ISSN 0002-9505. doi: 10.1119/1.1611479. URL https://doi.org/10.1119/1.1611479.

[43] Andrea Vitali. Microphone array beamforming in the pcm and pdm domain, 2018. URL https://www.st.com/resource/en/design_tip/dt0117-microphone-array-beamforming-in-the-pcm-and-pdm-domain-stmicroelectronic pdf.

[44] Mason Wang, Samuel Clarke, Jui-Hsien Wang, Ruohan Gao, and Jiajun Wu. Soundcam: A dataset for finding humans using room acoustics, 2024. URL https://arxiv.org/abs/2311.03517.

[45] BBC wildlife. What exactly is echolocation and how does it work? the incredible adaptation that allows animals to 'see' and communicate in the dark, 2025. URL https://www.discoverwildlife.com/animal-facts/what-is-echolocation.

[46] Junhyuk Woo, Hyeongmo Kim, Soon Ho Kim, and Kyungreem Han. Echo state property upon noisy driving input. *Complexity*, 2024(1):5593925, 2024. doi: https://doi.org/10.1155/2024/5593925. URL https://onlinelibrary.wiley.com/doi/abs/10.1155/2024/5593925.

[47] Rikiya Yamashita, , Rikiya Yamashita, Richard Kinh Gian, and Kaori Togashi. Convolutional neural networks: an overview and application in radiology. In *nsights Imaging*, page 611–629. springeropen, 2018. ISBN 978-0-12-822904-0. doi: https://doi.org/10.1007/s13244-018-0639-9. URL https://insightsimaging.springeropen.com/articles/10.1007/s13244-018-0639-9#citeas.

[48] Zhoutong Zhang. *Inferring Shape and Material from Sound*. Mtech thesis, MIT, 2021. Available at https://dspace.mit.edu/handle/1721.1/139579.